

BOX ARITHMETIC (Draft 1)

A finite multiset foundation
for mathematics

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Introduction

This book introduces a new orientation towards pure mathematics, called *Box Arithmetic*. The aim is to identify the essential objects, operations, and computational procedures — which we call *recipes* — needed for a broad foundation of the subject, and to build these from explicit inductive definitions with concretely decidable equality, rather than from existential or non-constructive assertions. The intended audience is serious students of mathematics and professionals already acquainted with the standard undergraduate curriculum.

Box Arithmetic connects with earlier traditions in mathematics and offers, for a significant and important domain, a more computationally transparent foundation than the currently dominant Zermelo-Fraenkel set theory with Choice (ZFC). ZFC rests on existential axioms asserting the existence of infinite sets, and its basic notion of equality — extensional equality of sets — is in general not decidable by a finite procedure. Box Arithmetic, by contrast, works entirely with finite objects whose equality is always decidable by an explicit algorithm. This is a genuine foundational difference, and for finitary mathematics — combinatorics, number theory, algebra, and the rudiments of geometry and linear algebra — we believe it represents a cleaner and more honest starting point. Whether this framework can be extended to cover the full scope of classical mathematics, including real analysis and general topology, is a question we take seriously and return to briefly in the conclusion. The primary notion is that of a *multiset*: a concretely specified, necessarily finite collection in which order is unimportant and repetitions are allowed and counted. A *set* is an important special case, as is a *list*.

This new framework also incorporates the Old Babylonian approach to "nothing" as an explicitly acknowledged absence — distinct from zero — a distinction that turns out to be essential when we introduce the transitive product in Section 10. The framework accords naturally with the ancient Greek orientation towards number as multiplicity, and is fully compatible with modern computation in a precise sense: every box can in principle be encoded as a finite string, equality of boxes is checkable by a terminating algorithm, and every operation defined in this paper is effectively computable. Explicit computational procedures — which we call *recipes* — play a central role throughout, and we regard their constructive character as a foundational feature rather than a convenience.

The primacy of natural number arithmetic via multisets

The book is divided into four Parts. The first Part deals with the beginnings of box arithmetic, and focussing on the most direct form of the theory, which is boxes built from natural numbers, and a corresponding arithmetic of unordered structures. The second Part introduces ordered structures of lists and corresponding vexels and maxels which brings in many concepts from linear algebra, combinatorics and algebra, but still over essentially the natural numbers. The third Part introduces negatives and subtractions by way of *anti-objects* or *opposites* — formal objects that cancel their counterparts in a multiset context — that carries a structural analogy to Dirac's introduction of anti-particles in quantum theory. That development opens up a richer and more involved algebraic world. The fourth Part attempts to show how this finite computationally based foundation can be used to form alternative frameworks for various core topics in mathematics. However the applications to analysis will await the book on Algebraic Calculus being written with Anna Tomskova.

The division between natural number and integer arithmetic, which is essential the divide between Parts I and II, deserves some explanation.

Many fundamental aspects of mathematics are all ready present in the more elementary and historically grounded theory that rests on the natural numbers alone. The oldest mathematical records we have are tallies and accounts — grain stored, livestock counted, debts recorded. The Sumerian accountant tracking numbers of sheep and goats does not first ask whether any two sheep or two goats are identical, and remove the duplicates. He counts multiplicities, because multiplicities are the point. The entire practical tradition of ancient arithmetic — Babylonian, Egyptian, Greek — is built on collections in which repetition is not only allowed but central, and these multiplicities are invariably positive.

When Euclid, in Book VII of the Elements, defines number, he writes:

A unit is that by virtue of which each of the things that exist is called one.
A number is a multitude composed of units.

This is much closer to a multiset conception than to a set-theoretic one. The move to sets as fundamental objects is historically a late intervention, emerging from the logicist and set-theoretic programs of Cantor, Frege, and their successors in the late nineteenth century — motivated by the need to handle infinite cardinalities and to sidestep the paradoxes of naive set theory — considerations that have no bearing on whether sets or multisets form the more natural foundation for finite arithmetic.

Compare Euclid's notion of (natural) number with the encoding adopted in modern set-theoretic foundations, due to von Neumann, in which natural numbers are defined to simultaneously serve as ordinals — each number being the set of all smaller numbers. This encoding is ingenious for the purpose of unifying cardinal and ordinal arithmetic within ZFC, but as a definition of number it is far removed from anything a student, an engineer, or an ancient mathematician would recognize: in terms of the

empty set $0 = \emptyset = \{\}$ we have the first natural $1 \equiv \{\emptyset\}$, $2 \equiv \{\emptyset, \{\emptyset\}\}$ and $3 \equiv \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ so that even the seemingly elementary number 4 is in reality

$$4 \equiv \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}\}.$$

Is this really a workable start to arithmetic?? Anyone other than a pure mathematician would probably think not.

Counting bracket pairs, the number n in this encoding requires 2^{n+1} braces — exponential in n . This is not merely a notational inconvenience: it reflects a genuine mismatch between the set-theoretic substrate and the objects being encoded.

The multiset approach immediately grounds us on a more sensible historical path. We will introduce natural numbers, following not just Euclid, but actually also the Sumerian practice long before the Greek period, where digits were created just by repeating a single stroke, and introducing a second symbol to denote a fixed multitude of such strokes. However we adopt the standard Hindu Arabic symbols, and names, for natural numbers. The basic operation of *addition* applies to all boxes, and when restricted to natural numbers supports counting, linear ordering, and multiples and linear combinations, and hence the rudiments of linear algebra at a very elementary level.

The arithmetical operations of Box Arithmetic form a natural hierarchy, each level defined by a uniform schema applied to the previous: addition gives multiplication, multiplication gives the caret operation, and so on. Crucially, none of these operations are intrinsically binary — they act on arbitrarily many boxes simultaneously. The successor operations beyond the caret appear to us as genuinely novel, and their full algebraic theory remains to be developed; we introduce them here primarily to establish that the hierarchy exists and that many aspects of arithmetic that we associated with the usual addition/multiplication duo have genuine extensions that go much further.

Polynomials — which we call *polynumbers* — are defined as multisets of natural numbers rather than as functions or formal expressions involving an abstract variable; this grounds them as explicit finite objects with decidable equality. Polynumbers in several variables, called multinumbers, fit into a natural hierarchy of levels, with natural numbers at level 1, polynumbers at level 2, and multinumbers at level 3. Classical results such as the Distributive laws, Inclusion-Exclusion, and the Pigeonhole principle extend naturally to this multiset setting, often in stronger box-level forms that imply the classical size-level versions.

Ordered structures — *lists* and *ordered sets* — are defined within Box Arithmetic in a way that scales naturally to n -tuples without the combinatorial overhead of iterated Kuratowski pairs. Specifically, where ZFC must encode an ordered n -tuple by nesting binary pairs, giving structures that grow quadratically in complexity with n , our list encoding is only linear in n . An immediate application of 2-lists is to introduce *proportions*: pairs of boxes considered up to a natural equivalence, giving affine ratios. This allows quotient arithmetic — positive fractions, ratios of polynumbers, ratios of multinumbers — to be introduced uniformly across the entire hierarchy of box objects.

In this paper we develop only this simplest case of proportions, and work throughout with natural number coefficients.

The objects which we call *unixels* and *pixels*, essentially 1-lists and 2-lists give rise, when collected into boxes, to *vexels* and *maxels*: generalizations of vectors and matrices that require no fixed dimension and no field of scalars. A novel non-commutative multiplication on pixels, the transitive product, uses the Babylonian notion of "nothing" to encode compatibility of composition — two pixels compose only when their inner entries match, and otherwise yield nothing rather than zero. This single idea, developed in Section 10, turns out to underlie matrix multiplication, function composition, and the definition of finite groups, all within a uniform framework. Maxels also provide natural encodings of graphs, relations, posets, and equivalences, making combinatorics a direct part of the box-arithmetic landscape rather than a separate subject.

There is a long-standing view that mathematics is discovered rather than invented. If one takes this seriously, we should expect that foundations may also be discovered if searched for. ZFC was developed to handle the "infinite sets" ostensibly required for analysis, while avoiding the paradoxes of naive set theory, but these motivations have little bearing on what the natural primitives for arithmetic and algebra should be. We hope to present here an architecture that feels less like a construction and more like a recognition: that finite multisets, explicit constructions, and decidable equality are not limitations to be overcome but the natural ground on which mathematics stands. This view is reinforced by modern computation: every object in Box Arithmetic is finitely representable, every equality check terminates, and every operation is effectively computable — properties that ZFC, designed before the computer age, was never required to satisfy. The natural framework for mathematics, as many computer scientist and programmers may have long suspected, resides in structures essentially belonging to those disciplines.

Part I

Natural number boxes

Objects, Types, Recipes and Proofs

1.1 Introduction

A foundational account of mathematics ought to specify reasonably clearly *what* the main objects are, how we *organize* them usefully into different kinds of objects, how we *manipulate* them, and what it means to *establish a result* about them. We thus identify four fundamental ingredients: *objects*, *types*, *recipes* and *proofs*. A prior ingredient, the *algebra of Boole*, provides the language for conditions and logical connectives and has at least at first a seemingly separate character. We now introduce each in turn, keeping them carefully separate from one another. We also acknowledge that these definitions are of an initial kind, and look forward to more careful and complete theories that clarify the notions here. Our suspicion is that such theories ought to be closely connected with areas of computing, programming and computer science.

Ultimately we want objects, types, recipes and proofs that can put written down explicitly and concretely using a fixed language of prior symbols. For convenience we take these symbols to be the standard alphanumeric and punctuation characters in common mathematical use. These may be strung together in the usual linear fashion to create (finite) strings, which we call generically **expressions**. Ultimately all the ingredients of mathematics that we will be considering can be reduced to expressions and relations between them.

Objects

Object A **mathematical object**, or simply an **object**, is a finite expression together with a specified decidable equality test. Thus two expressions may determine

the same object when the equality test identifies them.

The central objects of this work are **boxes**. Their definition and properties are developed from the ground up in the chapters that follow. For now we note only that boxes are explicitly writable, finitely described, and that equality of boxes is always decidable.

Other mathematical traditions introduce objects via axioms—assumed properties of abstract entities. We do not follow this path. Every object in our framework is explicitly constructible and finitely described. However we want to distinguish between different kinds of objects. For this we use the terminology of *types*.

Types

Type A **type** of object is a finite deterministic recognition procedure which, when applied to an object, terminates with output **Y** or **N**. The names of types will be in bold. We write $x \vdash \mathbf{T}$ to mean that the object x is recognized as having type **T**. It might be that a particular type may only apply to objects of a certain type, so we have types of types and so on. Refined types are indicated by a colon: thus $x \vdash \mathbf{T} : \mathbf{U}$ means that object x has type T and amongst object of type **T** additionally has type **U**.

For example, most objects in this book are of type **box**. So if A is a box then we write $A \vdash \mathbf{box}$. Natural numbers are particular types of boxes, their type will be denoted **Nat**. So to denote that m is a natural number, we may write $m \vdash \mathbf{box} : \mathbf{Nat}$, or if the context is clear $m \vdash \mathbf{Nat}$. When we list multiple types this way, we will begin with the larger, outermost types.

For example, once natural numbers have been introduced, the type **prime** for natural number could be introduced: any natural number m receives either **Y** (it is prime) or **N** (it is not). The assignment is finite and deterministic—running the primality test on any given natural number terminates and gives a definite answer. Thus prime numbers could be denoted by the type **box : Nat : prime** or if the context is clear, **Nat : prime** or just **prime**. However it could be that we also use the term prime as a type of different object, for example we could have prime polynumbers, denoted by the type **box : Poly : prime**, **Poly : prime** or just **prime**.

It might be that a type only has a finite number of objects. We call this a **finite type**. For example the type **Nat : prime : even** has only one object.

Several points deserve emphasis. *Types are not objects*. The values **Y** and **N** are not boxes; they belong to a separate logical infrastructure discussed in Section . A type is a one-way bridge from the world of objects to this infrastructure. *Types are also separate from recipes*. A type assigns to an object either **Y** or **N**; a recipe (Section) produces a new object. These are different activities, even though both are finite and deterministic.

Definition If S and T are types on the same kind of object, we say S is **contained in** T , written $S \leq T$, when any object assigned **Y** by S is also assigned **Y** by T .

Establishing such a containment generally requires a proof, not merely a computation.

Recipes

In mathematics we want to apply procedures to create new objects from existing ones. There are many current terms in mathematics that convey this idea, including prominently function, functional, map or mapping and transformation, while computer science has its own terminology, including program, routine and algorithm. However even in computer science there are subtle issues about precise definitions and notions of equality between algorithms. As a result, we use the term procedure informally here, sidestepping foundational questions about algorithms, but acknowledging the fundamental role that such primary questions have also in the foundations of mathematics.

We will be using the terms function, functional and transformation in box arithmetic in very particular mathematical senses. Hence we introduce the new term recipe to indicate the more largescale meta concepts that applies across types.

Recipe A **recipe** is a finite deterministic procedure that takes one or more objects of specified types as input, and produces an object of a (possibly different) specified type as output. A recipe must terminate on every valid input and produce a definite result. We will generally adopt the convention that the recipe R applied to the object A is denoted by $R[A]$. But some recipes, such as the fundamental operations of arithmetic, will have special notation.

Recipes are the workhorses of our arithmetic. Addition, multiplication, and the higher operations of Box Arithmetic are all recipes in this sense. A procedure that does not terminate on some input, or that produces an indeterminate output, is not a recipe in our framework.

Types and recipes are kept separate by design. A type assigns a meta-level value to an object; a recipe produces another object. Neither activity reduces to the other.

1.2 On the Nature of Proofs

We now come to the delicate issue of proofs in mathematics. Mathematical proofs as they appear in modern journals occupy a very wide spectrum of explicitness and checkability. At one end sit fully mechanical proofs, where every step is a finite computation and every term is defined. At the other end—where most of the literature lives—sit arguments whose steps require substantial expertise, unstated background, and a tacit agreement among readers not to ask too many questions.

This tacit agreement is sometimes harmless, as experienced readers may share enough context that a compressed argument conveys genuine content. But it can also conceal

serious difficulties. Consider an argument that begins: *Let M be a smooth manifold.* A smooth manifold is an infinite set equipped with infinitely many open sets satisfying infinite union and intersection conditions, covered by infinitely many charts into vector spaces whose points are tuples of real numbers—themselves constructed from infinite sets satisfying further infinite conditions. The convention is to agree that all of this makes sense and proceed. The level of implicit assumption is substantial, and it compounds with each additional layer of structure built on top.

We are not in a position to abolish this convention across mathematics. But we can be clear about what we are doing in the present work, and about what an honest attitude towards proof would look like.

Proofs as checkable computations

The ideal, which admittedly is seldom achieved, is that a proof is a finite object that can be checked step by step. Every term appearing in a proof should be defined somewhere, accessible in principle via a hyperlink. Every external result invoked should be named and linked to. The full logical structure of the argument should be traversable by a reader willing to follow every branch. In particular, a careful reader equipped with access to all definitions and prior results should be able to follow every proof, because every step is an explicit checkable operation.

This corresponds to a view of **proof as computation**: a proof is a finite sequence of recipe-applications, each step taking previously established objects and producing a new one, with the final step producing the desired conclusion. Checking such a proof requires only running the recipes and verifying the outputs—a finite decidable procedure at every step.

This is the central insight of the **Curry-Howard correspondence**, developed in theoretical computer science, which identifies propositions with types and proofs with programs. Computer proof assistants such as Coq, Lean and Agda implement this ideal: proofs are first-class objects, explicitly constructed and mechanically checkable. The difficulty with those systems, from our perspective, is not the checkability ideal but the assumptions they import—axioms guaranteeing completed infinite sets, real numbers, and other objects having no finite construction. Adapting the Curry-Howard programme to a setting that is fully finitary, fully decidable, and entirely assumption-free is a substantial undertaking, and the natural direction for the logical foundations of Box Arithmetic. We do not attempt to carry it out here; in what follows, proofs are presented in the ordinary mathematical way, with the understanding that a more formal account along these lines lies in the background.

So we will informally accept the following.

Definition A proof of a statement S is a finite, communicable argument that a competent reader can verify step by step, using the verification criteria associated to S .

The horizon

A further issue, rarely discussed explicitly, is whether a procedure invoked in a proof can actually be carried out within the framework in which one is operating.

Suppose a proof says: *raise x to the power n* . In principle this is a recipe, finite and deterministic. But if n is astronomically large—say, a tower of exponentials—then the output, though finite in principle, may be far too large to write down, store, or use in any further computation. The recipe terminates in theory but not in any practical sense.

Horizon The **horizon** is the boundary of what is genuinely reachable by the procedures available within a given framework.

A proof that invokes a computation beyond the horizon is not wrong in a logical sense, but it is not fully honest—it claims to construct something that cannot actually be constructed. The horizon is not a fixed line; it depends on the framework, the available machinery, and sometimes on physical constraints. What matters is that it be acknowledged.

In Box Arithmetic we are attentive to the horizon. The operations we define are finite and explicit, and the objects they produce are genuinely constructible. When we invoke a recipe, we mean that it can actually be run.

1.3 The Algebra of Boole logical framework

The algebra of Boole provides the simplest possible framework for conditions and logical reasoning: just two values **Y** and **N**, representing "Yes" and "no" combined via AND, OR, NOT and implication. It is logically prior to everything else in this work — simpler than arithmetic, simpler than boxes — and serves as the language in which type-assignments and logical connectives are expressed. Box arithmetic never produces **Y** or **N**; the algebra of Boole never produces boxes. The only connection is the type-assignment recipe, which runs in one direction only: from objects to types.

Note that we avoid the traditional role of "true" and "false". For us these notions are too laden with philosophical underpinnings. We will generally not be interested in whether or not a particular statement is "true": for us the relevant question is invariably: is a particular statement well defined, that is symantically meaningful, and if it is, is it correct or not. Thus we prefer not to say that $2 \times 3 = 6$ is true, but rather that it is correct.

While the Algebra of Boole originally has a prior level, elementary level of arithmetic, we will eventually see that it connects with structures inside box arithmetic.

1.4 Summary

We work with four kinds of things, kept carefully distinct:

Ingredient	Level	Character
Objects	Arithmetic	explicitly writable, decidable equality
Recipes	Arithmetic	finite deterministic, object to object
Types	Bridge (arithmetic $\rightarrow$ prior)	finite deterministic, object to $\{\mathbf{Y}, \mathbf{N}\}$
Boolean ops	Prior	AND, XOR, NOT and 13 derived connectives

Proofs are finite checkable objects built from recipe-applications. The Curry-Howard correspondence provides the right conceptual background: propositions correspond to types, and proofs correspond to recipes transforming proofs of hypotheses into proofs of conclusions. Existing implementations import infinitary assumptions we do not accept; adapting the framework to a fully finitary and assumption-free setting remains an important open direction.

We also insist on attention to the **horizon**: every procedure invoked in a proof should be genuinely executable within the framework, not merely terminating in principle. In subsequent chapters we build on this architecture, taking it as understood that all objects are explicitly constructible boxes, all types are decidable, all recipes terminate, and all claimed computations lie within the horizon.

The Algebra of Boole

2.1 Reasoning before mathematics

Before we can reason about mathematical objects—numbers, boxes, polynomials—we need a framework for reasoning itself. In the logical order of foundations, propositional logic comes first: it is the calculus of ANDs, ORs and NOTs that underlies all formal argument, and it can be developed without reference to any specific mathematical objects at all.

The formalization of logical reasoning has a history running parallel to, not prior to, mathematics. The Babylonians were solving quadratic equations and computing square roots centuries before Aristotle systematized the syllogism. But they were not, as far as we know, concerned with consistent abstract proofs of general statements. That particular development seems to have been a consequence of the ancient Greek genius. This concern systematizing reasoning, whether it be for mathematics or reasoning more generally begins in earnest perhaps with Aristotle's *Prior Analytics*, developed around 350 BCE. He asked not just whether a conclusion was true, but whether it followed necessarily from its premises, independent of the subject matter. That question—the question of valid inference—is what propositional logic addresses, and it has a primary role when applied to the statements of mathematics.

Aristotle's syllogistic was a theory of valid inference between categorical statements. Its paradigm example was the argument:

All men are mortal. Socrates is a man. Therefore Socrates is mortal.

From our perspective, this is a statement about types, albeit about real world object and not mathematics ones: the type **Man** is contained in the type **Mortal**, and Socrates belongs to the type **Man**, therefore Socrates belongs to the type **Mortal**.

Aristotle catalogued the valid forms of such arguments without the benefit of algebraic notation, but the underlying structure is what we might now call reasoning about type containment. Notice that the argument's validity requires no knowledge of the full extent of either type—no enumeration of all men or all mortals—only the containment relationship between them and Socrates's membership in one. The inference is local and finite; the potential vastness of the types behind it is simply irrelevant.

For nearly two thousand years Aristotle remained the standard framework for logic. Then, in the seventeenth century, Leibniz dreamed of something more ambitious: a *calculus ratiocinator*, a universal method of reasoning that would reduce disputes to calculation. In his *De Arte Combinatoria* (1666) he imagined that two philosophers disagreeing could simply sit down and compute who was right. The dream was not realised in his lifetime, but it pointed in exactly the right direction.

The algebraic realisation came in 1847, when George Boole published *The Mathematical Analysis of Logic*. Boole's central idea was to represent logical propositions as algebraic expressions, and logical reasoning as algebraic manipulation. He used $+$ and $\times$ as his operations, with the special property that every variable x satisfies $x^2 = x$ —a consequence of the fact that logical variables take only the values 0 (false) and 1 (true). In the same year, Augustus De Morgan published *Formal Logic*, taking a rather different, more set-theoretic direction.

Both authors were trying to systematise the same thing: the kind of reasoning Aristotle had begun. The approaches diverged, and it was De Morgan's direction that prevailed in subsequent development. John Venn gave it a vivid diagrammatic form; Edward Huntington gave it a clean axiomatic treatment in 1904. The result—three operations AND, OR, NOT, governed by the laws of a lattice—is what is today called Boolean algebra, and what is taught in engineering and computer science courses worldwide.

We take a different path. We return to Boole's algebraic orientation, grounding it in a particularly simple and finite arithmetic which ultimately turned out to be exactly the arithmetic of the bifold, or finite field with 2 elements. The result is the *Algebra of Boole*—not Boolean algebra—and it has a decisive advantage: every propositional expression has a unique canonical polynomial form. This makes logical reasoning genuinely algebraic and algorithmic. The full development is given in Chapter ??; here we introduce only what is needed for the work ahead.

2.2 Logic as a foundation for mathematics

Classical foundations for mathematics—set theory in the style of ZFC—do not begin with sets. They begin with first-order logic: a language with variables, quantifiers (for all, there exists), predicates and connectives. Sets are then introduced on top of this logical substructure.

This ordering reflects a genuine philosophical point: logic must be in place before mathematical objects can be discussed. But it also imports the full machinery of

first-order logic—in particular, quantification over infinite domains—before a single mathematical object has been defined. For the classical mathematician this is unproblematic; for us it is not.

Our foundations take a different form. The logical substructure we need is simpler and more elementary: it is propositional logic, the logic of types without quantifiers. If we are talking about a finite set of people, we can introduce types **Tall** and **Smart** and **Athletic**, and then can discuss **Tall AND Smart**, **NOT Old**, **Smart IMPLIES Athletic**. We can ask whether one type is contained in another for specific objects, or reason with finitely many. This is the level at which the Algebra of Boole operates, and it is completely finite and decidable.

The step to open-ended reasoning—are all primes odd? does there exist an even prime greater than 2?—requires more: it requires types in the sense of the previous chapter, and proofs in the sense described there. But the algebra of propositional logic for finite types, handled algebraically, comes first, and allows us to establish the main laws. It is the logical substructure prior to arithmetic.

2.3 Types instead of sets

In conventional Boolean algebra, the logical connectives AND, OR and NOT correspond to the set-theoretic operations of intersection, union and complement. The Venn diagram makes this vivid: the AND of two types is the region where both circles overlap; the OR is the region covered by either; the NOT is everything outside.

This grounding in set theory is natural but carries a cost: it inherits infinitely many elements, membership as a primitive relation—and it loses canonical form. De Morgan's laws say that the complement of an AND is the OR of the complements, and vice versa. This means the same propositional expression can be written in many different ways, and there is no algorithmic procedure for deciding whether two expressions are equivalent without case analysis.

We replace sets with types. A type is a decidable property of objects—a finite deterministic procedure returning **Y** or **N**, as introduced in the previous chapter. Types do not presuppose infinite collections. They are concrete and constructive. And the arithmetic of **Y/N**—the prior level we already have—is exactly the arithmetic needed to reason about them.

The Algebra of Boole is the calculus of types. It tells us how to form new types by combining old ones (AND, OR, NOT), how to ask whether one type is contained in another, and how to find the canonical form of any propositional expression built from types.

2.4 The two-element arithmetic

The values **Y** and **N** form a complete arithmetic of their own. Setting $\mathbf{Y} \equiv 1$ and $\mathbf{N} \equiv 0$, the two primitive operations are AND (multiplication) and XOR—exclusive or—(addition). All other logical connectives are derived from these two:

$$\text{NOT}(a) \equiv 1 + a, \quad \text{OR}(a, b) \equiv a + b + ab, \quad (a \rightarrow b) \equiv 1 + a + ab.$$

Two properties distinguish this arithmetic from ordinary arithmetic and make propositional reasoning finite and decidable:

$$x + x = 0 \quad (\text{Boole additivity}), \quad x^2 = x \quad (\text{Boole multiplicativity}).$$

Both hold by direct inspection. Together they ensure that any propositional expression, however written, reduces by routine algebra to a unique canonical form—a multilinear polynomial over $\{0, 1\}$ with no repeated variables. This is the central fact of the Algebra of Boole:

Checking equivalence requires no ingenuity, only arithmetic. The full account—including the bifold tables, the 16 binary connectives, the Boole-Möbius transform, and applications to circuit analysis—is given in Chapter ??.

2.5 A finite example

To ground these ideas concretely, consider four people—Alice, Bob, Carol and Dave—each satisfying or not satisfying four types:

Type	Alice	Bob	Carol	Dave
Tall T	1	0	1	1
Smart S	0	1	1	0
Athletic A	1	1	0	1
Old O	0	0	1	1

Each type is a binary vector, and logical operations act componentwise. The question *are all smart people athletic?* becomes: does $S \times (1 + A) = 0$? Computing:

$$(0, 1, 1, 0) \times (0, 0, 1, 0) = (0, 0, 0, 0).$$

Yes—every smart person is athletic. The question *is there someone old but not smart?* becomes: does $O \times (1 + S) \neq 0$? Computing:

$$(0, 0, 1, 1) \times (1, 0, 0, 1) = (0, 0, 0, 1).$$

Yes—Dave is old but not smart.

Every question of the form *does every X satisfy Y?* or *does some X satisfy Y?* reduces to checking whether a componentwise computation yields the zero vector or not. Logical questions become arithmetic questions.

2.6 Propositional logic in this book

The Algebra of Boole handles propositional reasoning—reasoning about types—completely and honestly at the finite level. When the domain is finite, every logical question is answered by direct computation, as in the example above. When the domain is open-ended—all natural numbers, all boxes—the same logical structure applies, but answering the question requires a proof in the sense of the previous chapter: a finite constructed argument establishing the containment for all relevant objects.

This is where propositional logic ends and the arithmetic of Box Arithmetic begins. The Algebra of Boole is its logical substructure—historically grounded, algebraically clean, and completely decidable within its domain. The full computational machinery is developed in Chapter ??.

Boxes and their addition

The objects we work with — boxes — are hereditarily finite multisets: finite collections, allowing repetition, whose elements are themselves finite collections of the same kind. Structures of this type have appeared in mathematical logic in the context of hereditarily finite sets (where repetition is not allowed), studied by Ackermann and later by Kirby among others. Our departure is to allow multiplicities, making multisets rather than sets the primary notion, and to insist that every construction be explicitly finite and every equality decidable by a terminating algorithm. The formal definition is inductive, starting from arguably the simplest data structure.

3.1 The basic objects are finite multisets: boxes

Definition The **empty box** is the expression $\square$. We denote it also by the expression $\llbracket \rrbracket$ involving a left lower square bracket, a space, and a right lower square bracket, and give it also the symbol 0 and the name **zero**.

We may introduce the type **Zero** for the empty box. All other boxes are defined inductively, starting with empty boxes.

Definition A **box** is either $\square$, or an expression of the form $M = \llbracket A, B, \dots, K \rrbracket$ consisting of previously defined boxes $A, B, \dots, K$ separated by commas, enclosed in square lower brackets, with the understandings that *order does not matter*, and *repetitions are allowed*. We may introduce the type **Box** for a box.

If $M = \llbracket A, B, \dots, K \rrbracket$ then the boxes $A, B, \dots, K$ are then the **elements** of the box M . If the box X is an element of the box Y , then we write $X \in Y$. Since every box is

built from previously defined boxes, no box can contain itself as an element, either directly or through a chain of containment. This well-foundedness is a direct consequence of the inductive construction and requires no separate assumption — in contrast to ZFC, where the axiom of foundation (regularity) must be explicitly imposed to rule out sets containing themselves. Note that we will reserve left and right lower square brackets $[$ and $]$ exclusively to denote boxes.

Example 3.1 We have already defined $\square = 0$ to be a box. Thus $[\square]$ is a box, which we call **one**, and denote with the symbol $1 = [0]$. Another example of a box is $[\square, \square] = [0, 0]$ which we call **two**, and denote with the symbol 2 . Thus $2 = [0, 0]$. The box $[\square, \square, \square] = [0, 0, 0]$ we call **three**, and denote it with the symbol 3 . Thus $3 = [0, 0, 0]$.

Our representation of empty boxes with spaces, and the use of commas to separate elements, are conveniences for human viewers, but they are not logically necessary, as the pattern of left and right lower square brackets completely determines the elements.

Remark 1 If we write the empty box $0 = \square$ as $\square$, then a general box can be viewed as a sequence of lower left and lower right brackets, together with spaces and commas. Strictly speaking the spaces and commas are not logically necessary, and by dispensing with them, any box can be written as a bracket string satisfying the *Catalan condition*: reading left to right, the count of left brackets never falls below the count of right brackets, and the two totals are equal. However, since boxes are unordered, different bracket strings may represent the same box — reordering elements inside any box yields a different string but an identical box. The number of distinct bracket strings of length $2n$ representing a single box is the $(n - 1)$ -th Catalan number C_{n-1} , but the number of distinct boxes using exactly n bracket pairs is the n -th term of the sequence

$$1, 1, 2, 4, 9, 20, 48, 115, 286, 719, 1842, \dots$$

— unlabeled unordered rooted trees with n nodes (OEIS A000081), which grows considerably more slowly. For small n we can verify these values. For $n = 1$ we have just the box 0 . For $n = 2$ just the box $1 = [0]$. For $n = 3$ the two boxes $2 = [0, 0]$ and $[1]$. For $n = 4$ the four boxes $3 = [0, 0, 0]$, $[0, 1]$, $[2]$ and $[[1]]$. And for $n = 5$ the nine boxes are organized by the partitions of the four interior pairs: $[0, 0, 0, 0]$, $[0, 0, 0, 1]$, $[1, 1, 1]$, $[2, 0]$, $[0, [1]]$, $[3]$, $[[0, 1]]$, $[[2]]$ and $[[[1]]]$. The gap between these two counts is exactly what makes equality of boxes non-trivial: two bracket strings represent equal boxes not when they are identical as strings, but when their corresponding unordered rooted trees are isomorphic. We do not use the Catalan characterization as our primary definition, since the Catalan condition itself presupposes counting; but it provides a useful alternative viewpoint, and the A000081 count gives a measure of the genuine complexity of the box universe at each level.

3.2 Equality of boxes

Definition Equality of boxes is defined inductively. Any two empty boxes are defined to be equal. Two non-empty boxes A and B are equal precisely when there exists a bijection between their elements — counting multiplicities — under which paired elements are equal boxes. Since every box is finite and each step of the comparison moves to strictly smaller boxes, this procedure always terminates.

In the language of graph theory, a box corresponds naturally to an unordered rooted tree, with the outermost box as root and its elements as children. Equality of boxes is then exactly unordered rooted tree isomorphism, for which efficient algorithms are known. This is worth emphasising from a foundational standpoint: equality of boxes is not an axiom but a decidable relation, established by a concrete terminating procedure. This contrasts with ZFC, where extensional equality of sets — two sets are equal if they have the same members — is an axiom of the theory, with no associated decision procedure. It also aligns Box Arithmetic with the tradition of definitional equality in constructive type theory, notably Martin-Löf Type Theory, where the fundamental equality of the system is likewise decidable by a normalization procedure. In practice, for the explicit small boxes that appear throughout this paper, equality is obvious by inspection; the foundational point is that it is always decidable in principle, with no appeal to non-constructive principles.

We will reserve the right to in the future augment our definition of boxes, that is to include other, related, objects as well.

Example 3.2 Since $0 = \square$ is a box, we have that $[0]$ is a box; we call this **one**, and denote it by the symbol 1. The box $[0, 0]$ is called **two** and we denote it by the symbol 2, and the box $[0, 0, 0]$ is called **three** and denoted by the symbol 3, and so on, using the standard Hindu Arabic names and symbols. Here is a more complicated box, showing that our naming of basic boxes allows different representations of the same box.

$$\begin{aligned} p &\equiv [\square, [[\square], [\square]], [\square, \square, \square], [\square, \square, \square]] \\ &= [0, [[0], [0]], [0, 0, 0], [0, 0, 0]] = [0, [1, 1], [0, 0, 0], [0, 0, 0]] = [0, [1, 1], 3, 3]. \end{aligned}$$

One of the challenges in the subject is organizing naming conventions for more complicated boxes.

3.3 Addition of boxes

The fundamental operation on boxes is **addition**. Addition of boxes is the mathematical equivalent of dumping all the contents of several physical boxes into a single new physical box.

Definition If $A, B, \dots, K$ are boxes, then the **sum** $A + B + \dots + K$ is the box consisting of all the elements of the boxes $A, B, \dots, K$, again with repetitions allowed. We note that addition is not intrinsically a binary operation.

Concretely: addition is the operation of emptying several boxes into a single new box. We emphasise that addition is not intrinsically a binary operation — any finite collection of boxes may be added simultaneously, and the result is independent of any grouping or ordering. This distinguishes box addition from set union, which discards multiplicities: the union of $[0]$ and $[0]$ as sets would give $[0]$ while their sum as boxes gives $[0, 0] = 2$.

Example 3.3 If $M = [a, a, K]$ and $N = [a, b]$ and $P = [K, L]$ are boxes, with a, K, b and L previously defined boxes, then
 $M + N + P = [a, a, K, a, b, K, L] = [a, a, a, K, K, b, L]$.

Example 3.4 Here is an example of adding several boxes:
 $[0, 0] + [[0, 0], 0] + [[0, 0, 0], [0]] = [0, 0, [0, 0], 0, [0, 0, 0], [0]] = [0, 0, 0, 1, 2, 3]$.

Addition of boxes is **commutative** — $A + B = B + A$ — since the definition places all elements into a single unordered collection, making the order of the summands irrelevant. It is **associative** — $(A + B) + C = A + (B + C)$ — since grouping is likewise irrelevant: in each case the result is simply the box of all elements of A, B and C together. The empty box 0 acts as the **additive identity**: $0 + A = A + 0 = A$ for any box A , since adding a box with no elements changes nothing. Together these properties say that boxes under addition form a "commutative monoid", with 0 as the identity element. For a genuinely *multi-input unordered operation*, commutativity and associativity are not additional properties requiring verification — they are already encoded in the definition, which specifies neither an order among the summands nor a grouping.

We now point out two not entirely obvious properties of addition of boxes, and give informal reasons why they are valid.

Proposition 1 (Cancellation property) If A, B and C are boxes, and $A + B = A + C$, then $B = C$.

Proof. If $A + B = A + C$ then the elements of $A + B$ can be paired off with equal elements of $A + C$. The elements of A in both $A + B$ and $A + C$ can be paired off as equal, which leaves the elements of B and C which can be paired off as equal. So $B = C$.

Proposition 2 (Zero sum property) If $A, B, \dots, K$ are boxes and $A + B + \dots + K = 0$, then we must have $A = B = \dots = K = 0$.

Proof. If $A + B + \dots + K = 0$ then this sum has no elements, so none of $A, B, \dots, K$ can have any elements, so that $A = B = \dots = K = 0$.

When we introduce integer arithmetic by way of boxes, this last rule will need to be modified.

3.4 Subboxes and the partial order

Addition of boxes also introduces an ordering on boxes.

Definition If A and B are boxes, we say that A is a **subbox** of B , written $A \leq B$ or $A \subseteq B$ (or equivalently $B \geq A$ or $B \supseteq A$) precisely when there is a box D such that

$$A + D = B.$$

By the Cancellation property, such a D is unique when it exists, in which case we call it the **complement** of A in B . This relation is precisely multiset inclusion: $A \leq B$ precisely when every element of A appears in B with at least as great a multiplicity.

This relation defines a **partial order** on boxes — reflexive, antisymmetric, and transitive. Reflexivity is immediate since $A + 0 = 0 + A = A$. Transitivity is straightforward. Antisymmetry requires both Propositions 8 and 9: if $A + D = B$ and $B + E = A$ then $A + D + E = A$ so $D + E = 0$ by cancellation, hence $D = E = 0$ by the zero sum property, giving $A = B$.

Boxes and rooted unordered trees

Overview

Rooted unordered trees provide a simple and important model for the objects in Box Arithmetic. Indeed a box may be viewed as a *rooted unordered tree*: the outer box is the root, and the elements of that box are its children. If an element is itself a box, then its elements give the next generation of children, and so on. Since the elements of a box are unordered, the corresponding tree is unordered. Since every box is finitely built from the empty box, the corresponding tree is finite and well-founded. Because trees have been extensively studied in computer science and combinatorics, it will be important to connect with this realization of boxes and their arithmetic, as we have ready made notation and algorithms that can be profitably employed.

This chapter reviews basic definitions for rooted unordered trees as concrete finite objects in their own right. Note that from a foundational point of view these definitions are not part of box arithmetic: they allow us to connect with an existing area of study. We begin with ordinary graph language, but quickly specialize to the rooted unordered case of trees. The terminology is useful because it gives an efficient geometric language for boxes: root, node, leaf, child, parent, ancestor, descendant, level, depth, height, branch, subtree, pruning and isomorphism. In a later Chapter on combinatorial applications of maxels we will return to definitions including that of graph and tree, but there within the box world. For this reason the definitions in this Chapter are here to clarify the connection between boxes and trees, and to allow us to transfer terminology. Hence they are in italics, not in bold as are the main definitions of the book.

The main point is that equality of boxes is the same as isomorphism of rooted unordered trees. This is not an axiom, but a finite comparison procedure. Two trees are equal when their roots have matching families of equal rooted subtrees, counted

with multiplicity. Thus rooted unordered tree theory gives both a visual model for Box Arithmetic and a practical algorithm for equality.

4.1 Graphs and trees

A *graph* consists of a finite set V of vertices, together with a finite set E of edges, where each edge connects two distinct vertices of V . In this book all graphs are finite, undirected and simple: there are no loops and no multiple edges. A graph may be pictured by drawing a dot for each vertex and a line segment for each edge. The drawing is not the graph itself; it is only a representation. The graph records which vertices are joined to which other vertices.

A *path* in a graph is a list of vertices $v_0, v_1, \dots, v_n$ such that consecutive vertices v_i and v_{i+1} are joined by an edge, and no vertex is repeated unless explicitly allowed. The *length* of the path is n , the number of edges traversed. A graph is connected precisely when any two vertices can be joined by a path. A *cycle* is a path of strictly positive length which starts and ends at the same vertex, with no repeated vertices apart from this start/end repetition.

A *tree* is a connected (non-empty) graph with no cycles. Equivalently, a tree is a finite graph in which any two vertices are joined by exactly one path. This uniqueness of paths is often the most useful way to think about a tree.

Rooted trees

A *rooted tree* is a tree T together with a distinguished vertex r , called the *root*. We write (T, r) , or simply T when the root is understood. In diagrams the root will always be the topmost vertex. The root turns an unrooted tree into a hierarchical object; since every vertex has a unique path from the root, it is meaningful to talk about above and below, ancestors and descendants, parents and children.

Let T be a rooted tree with root r , and let v be a vertex of T . The *depth* of v is the length of the unique path from r to v . The root has depth 0. The vertices at depth k are said to form *level* k of the tree.

If v is not the root, then the *parent* of v is the vertex immediately before v on the unique path from the root to v . If u is the parent of v , then v is a *child* of u . Vertices with the same parent are called *siblings*. The *children* of a vertex v are the vertices immediately below v . The number of children of v is the *arity*, *valency*, *branching number*, or *out-degree* of v . In this chapter the term branching number is often the most descriptive. A *chain* is a tree where no vertex has branching number greater than 1.

A *leaf* is a vertex with no children. A non-leaf vertex is called an *internal vertex*. The root may be a leaf; this happens precisely for the one-vertex tree. The *height* of a

rooted tree is the maximum depth of any vertex. Equivalently, it is the length of the longest path from the root to a leaf. The one-vertex tree has height 0. A chain of three vertices, rooted at one end, has height 2.

The *size* of a tree is its number of vertices. Sometimes it is also useful to count edges. For a finite tree with n vertices there are $n - 1$ edges.

Ancestors, descendants and subtrees

Let u and v be vertices in a rooted tree T . We say that u is an *ancestor* of v precisely when u lies on the unique path from the root to v . We say that v is a *descendant* of u if u is an ancestor of v . We allow a vertex to be its own ancestor and its own descendant. If u is an ancestor of v and u is not v , then u is a *proper ancestor* of v , and v is a *proper descendant* of u .

For a vertex v of T , the *principal subtree* at v , denoted T_v is the rooted tree consisting of v and all descendants of v , with v as root. Thus the principal subtree at the root is the whole tree. The principal subtree at a leaf is a one-vertex tree.

The *root path* to a vertex v is the unique path from the root r to v . The union of the root path to v and the principal subtree T_v is again a connected rooted subgraph of T . It consists of the path from the root down to v , together with every descendant of v . This object will be useful for us, but does not seem to have a universally standardized name in graph theory. In this book we shall call it the *rooted branch* of v . Thus the rooted branch of v records both how to reach v from the root and everything that lies below v .

[For Box Arithmetic, the principal subtree T_v corresponds to the box sitting at v . The branch through v corresponds to that box together with its complete address inside the larger box.]

Ordered and unordered rooted trees

A rooted tree as usually drawn on a page has a left-to-right order among the children of each vertex. This order may come from the drawing rather than from the tree itself. A *rooted ordered tree* is a rooted tree in which the children of each vertex are arranged in a definite order. A *rooted unordered tree* is a rooted tree in which the children of each vertex are not ordered. Only the multiset of children matters.

The distinction is important. For Box Arithmetic, rooted unordered trees are the natural objects of interest, because the elements of a box have no order. If $A = [B, C]$, then also $A = [C, B]$. The two children B and C may be displayed in either order without changing the box.

Rooted unordered tree isomorphism

Two rooted trees T and U are *isomorphic as rooted trees* precisely when there is a bijection from the vertices of T to the vertices of U which sends the root of T to the root of U and preserves adjacency and non-adjacency. For rooted unordered trees, no left-to-right order of children has to be preserved, since there is no such order. What must be preserved is only the parent-child structure.

Equivalently, two rooted unordered trees are isomorphic if their roots have the same number of children, and the rooted subtrees hanging from the children of the first root can be matched bijectively with the rooted subtrees hanging from the children of the second root, with matched subtrees themselves isomorphic.

This is a recursive definition of equality.

Recursive construction of rooted unordered trees

Rooted unordered trees may be constructed recursively in a way parallel to the construction of boxes. The one-node tree, denoted $\bullet$, is a rooted unordered tree consisting of a single root and no other vertices.

If $T_1, T_2, \dots, T_k$ are rooted unordered trees, not necessarily distinct, then $[T_1, T_2, \dots, T_k]$ is the rooted unordered tree whose root has as its children the roots of $T_1, T_2, \dots, T_k$, with each T_I hanging below its corresponding root. The order of $T_1, T_2, \dots, T_k$ is irrelevant. If $\bullet$, this is the one-node tree.

This is the same recursive pattern as boxes. The one-node tree corresponds to the empty box. A tree whose root has children $T_1, T_2, \dots, T_k$ corresponds to the box whose elements are the boxes corresponding to $T_1, T_2, \dots, T_k$.

4.2 The box / rooted unordered tree correspondence

To create a tree $T[A]$ from a box A , we proceed as follows: associate to the outside brackets of the box the root node. To every element of the box we create a child node of the root. And we continue recursively, until singleton empty boxes correspond to leaf nodes.

It is often convenient to use the following dictionary between boxes A and rooted unordered trees $T[A]$.

Box language	Tree language
$0 = \square$	one-node rooted tree
A is an element of B	$T[A]$ is a child-subtree of $T[B]$
number of elements of A	number of children of root of $T[A]$
box equality	rooted unordered tree isomorphism
box depth	tree height
multiplicity of an element	multiplicity of equal child-subtrees

Under this dictionary, boxes and rooted unordered trees are not merely analogous. They are essentially the same finite structure viewed in two different languages. In fact we will expand on this dictionary as we proceed.

4.3 Examples of small boxes and small rooted unordered trees

TO DO: create diagrams

Let us list the rooted unordered trees with small numbers of vertices. Since we ignore the order of children, there are fewer such trees than there are bracket strings or ordered rooted trees. Since

There is one rooted unordered tree with 1 vertex:

1. • In box notation $\square$

There is one with 2 vertices:

1. a root with one leaf child. In box notation $[\square] = 1$

There are two with 3 vertices:

1. a root with two leaf children, corresponding to $[\square, \square]$
2. a chain of length 2 below the root, corresponding to $[[\square]]$

There are four with 4 vertices, corresponding to $[\square, \square, \square]$, $[\square, [\square]]$, $[[\square, \square]]$, $[[[\square]]]$

There are nine with 5 vertices. They may be organized by the multiset of rooted subtrees hanging from the root. This is a good small exercise: every 5-vertex rooted unordered tree is obtained by distributing the remaining 4 vertices among the child-subtrees of the root, with order irrelevant.

4.4 Summary of terminology

root: the distinguished top vertex of a rooted tree.

node or vertex: one of the points of the tree.

edge: a connection between two adjacent vertices.

parent: the vertex immediately above a non-root vertex.

child: a vertex immediately below a given vertex.

siblings: vertices with the same parent.

leaf: a vertex with no children.

internal vertex: a vertex with at least one child.

depth of a vertex: the length of the path from the root to that vertex.

level k : the vertices of depth k .

height of a tree: the largest depth of any vertex.

ancestor: a vertex on the path from the root to a given vertex.

descendant: a vertex lying below a given vertex.

principal subtree at v : the rooted tree consisting of v and all descendants of v .

root path to v : the unique path from the root to v .

branch through v : the union of the root path to v and the principal subtree at v .

ordered rooted tree: a rooted tree in which children are ordered.

unordered rooted tree: a rooted tree in which children are not ordered.

isomorphism: a root-preserving adjacency-preserving bijection of vertices.

canonical name: a standard finite string or box expression representing the rooted unordered tree shape.

pruning: deleting all structure below a specified depth.

horizon: a declared finite bound on admissible size or complexity.

Concluding remarks

Rooted unordered trees are not an optional visualization for Box Arithmetic. They reveal the structural content of boxes. The empty box is the one-node rooted tree; the elements of a box are the child-subtrees of the root; equality of boxes is rooted unordered tree isomorphism; and the recursive construction of boxes is the recursive construction of rooted unordered trees.

This point has both practical and philosophical value. Practically, tree language gives efficient algorithms for equality, canonical naming, pruning and structural induction. Philosophically, it reinforces the central aim of Box Arithmetic: mathematics should be built from finite, concrete, explicitly recognizable objects, with equality and operations given by terminating recipes rather than by appeal to completed infinite domains.

Natural numbers, counting and ordering

Now let us turn to the simplest level of arithmetic in this new world once we go beyond the empty box 0.

Definition A **natural number** is a box of zeros: a box each of whose elements is the empty box $\square = 0$. We regard $\square = 0$ as an empty box of 0's, hence it is also a natural number.

The natural numbers $0, 1, 2, 3, \dots$ are thus the boxes $\square, [\square], [\square, \square], [\square, \square, \square], \dots$ — essentially a unary numeral system, where the number n is recorded as n tallies/primitive objects inside a box. As we shall see, these boxes satisfy the familiar Peano axioms for arithmetic, but derived here from the box structure rather than assumed as axioms.

5.1 Induction

The natural numbers are built in a single direction: starting from 0, each new natural number is obtained by adding one more zero. This directional, cumulative construction is what makes induction possible.

Theorem 1 (The Induction Principle) Suppose P is a property of natural numbers such that: $P(0)$ holds, and for any natural number k , if $P(k)$ holds then $P(k+1)$ holds. Then $P(n)$ holds for every natural number n .

Proof. By the well-foundedness of boxes (Definition 1), every natural number n is reached from 0 by exactly n applications of the successor operation. Since P holds at 0 and is preserved at each step, it holds at every finite stage of this construction, hence holds for n .

Remark 2 This is the only result in the paper whose proof appeals directly to the meta-level structure of the construction rather than to previously established box arithmetic. Its special foundational status is reflected in its designation as a Theorem rather than a Proposition.

We note that "property" here means a property whose truth or falsity for any given natural number is determinable by an explicit finite procedure — in practice, a recipe in the sense discussed before Definition 10, or a condition that any competent reader could in principle verify for each specific n .

n. The question of precisely which properties are admissible connects to deep foundational issues we leave for a subsequent treatment. All properties appearing in this paper are clearly of the required kind.

We may add $1 = [0]$ to any natural number n to obtain its **successor** $\text{succ}[n] = n + 1$. The ordering

$$0 < 1 < 2 < \cdots < n.$$

is established by this construction, and induction gives us the tool to prove properties about all natural numbers simultaneously.

This ordering agrees with the general inclusion partial order on boxes, and any natural numbers m and n are comparable so that either $m < n$ or $m = n$ or $n < m$. This gives us notions of "larger" and "smaller" for natural numbers.

5.2 Max and min

Definition For natural numbers m and n we define their **minimum** and **maximum** to be respectively the natural numbers

$$\min[m, n] \equiv \begin{cases} m & \text{if } m \leq n \\ n & \text{if } m > n \end{cases} \quad \max[m, n] \equiv \begin{cases} n & \text{if } m \leq n \\ m & \text{if } m > n \end{cases}.$$

These are well-defined because the ordering on natural numbers is total (Section 3.1): for any m and n exactly one of $m \leq n$ or $m > n$ holds.

Remark 3 These two recipes entirely encode the order structure on the natural numbers: knowing $\min[m, n]$ and $\max[m, n]$ for any m and n is equivalent to knowing the ordering. They also introduce two important algebraic structures. First, the natural numbers under $\min$ and $\max$ form a *distributive lattice* — the algebraic

structure underlying order theory. Second, replacing ordinary addition with $\min$ and ordinary multiplication with $+$ gives the *tropical semiring*, a structure of growing importance in geometry, combinatorics, and optimisation. Both structures emerge here over the natural numbers without any subtraction or division.

Proposition 3 (Max/min identities) For natural numbers m, n and p we have

$$\begin{aligned}
 \max[m, m] &= m & \min[m, m] &= m & (\text{Idempotent}) \\
 \max[m, n] &= \max[n, m] & \min[m, n] &= \min[n, m] & (\text{Commutative}) \\
 \max[m, \max[n, p]] &= \max[\max[m, n], p] & & & (\text{Associative}) \\
 \min[m, \min[n, p]] &= \min[\min[m, n], p] & & & (\text{Associative}) \\
 \max[m, \min[m, n]] &= \min[m, \max[m, n]] = m & & & (\text{Absorption}) \\
 \max[m, \min[n, p]] &= \min[\max[m, n], \max[m, p]] & & & (\text{Distributive}) \\
 \min[m, \max[n, p]] &= \max[\min[m, n], \min[m, p]] & & & (\text{Distributive})
 \end{aligned}$$

Proof. All of these identities follow directly from the definitions of $\min$ and $\max$ and the total ordering of natural numbers. The two-variable identities are verified by checking the cases $m \leq n$ and $m \geq n$. The three-variable identities — associativity and distributivity — are verified by considering all possible orderings of m, n and p ; since the ordering is total there are at most $3! = 6$ cases, and symmetry reduces the work further in each case.

5.3 Addition of natural numbers, sizes and multiplicities

Since natural numbers are boxes, they can be added. The result is again a natural number. We record this as our first closure result.

Proposition 4 The natural numbers are closed under addition: if m and n are natural numbers then $m + n$ is a natural number.

Proof. Both m and n are boxes of zeros. Their sum $m + n$ collects all elements of both, all of which are 0, giving a box of zeros.

Addition of natural numbers then gives the usual familiar operation, with $m + n$ equal to the natural number whose value is the ordinary sum.

Example 5.1 Here is a sum of two natural numbers:

$$3 + 4 = [0, 0, 0] + [0, 0, 0, 0] = [0, 0, 0, 0, 0, 0, 0] = 7.$$

Note that since addition of boxes is not intrinsically binary, neither is addition of natural numbers.

Example 5.2 Here is a sum of three natural numbers:

$$4 + 1 + 6 = [0, 0, 0, 0] + [0] + [0, 0, 0, 0, 0, 0] = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0] = 11.$$

Natural numbers are the counting numbers: they arise whenever we ask "how many?" The most fundamental such question in Box Arithmetic is how many elements a box has — its size. This is our first example of a **recipe** in Box Arithmetic: an explicitly defined procedure that produces a unique output from its input by a finite process that always terminates. All recipes in this paper are of this character: they are defined by explicit rules operating on the finite structure of boxes, and their termination follows either from the well-foundedness of boxes established in Definition 1, or from finite indexing. We will indicate recipes by using square brackets. We leave a fuller formal account of what classes of recipes are admissible to a subsequent foundational treatment.

Definition The **size** $s[A]$ of a box A is the recipe that replaces each element of A with the empty box 0 , discarding all internal structure. Equivalently, $s[A]$ is the unique natural number equal to the number of elements of A , counting multiplicities.

Proposition 5 The size of any box is always a natural number.

Proof. Replacing every element of A with 0 gives a box of zeros, which is by definition a natural number.

Example 5.3 Thus if $A \equiv [7, 7, [3, 5], [[4], 2]]$ then $s[A] = [0, 0, 0, 0] = 4$, and we say that A has 4 elements.

For any boxes $A, B, \dots, K$ we have

$$s[A + B + \dots + K] = s[A] + s[B] + \dots + s[K].$$

Definition If A and B are boxes, then the **multiplicity** $m_B(A)$ of B in A is the number of times B occurs as an element of A . We may define this as the size of the box obtained from A by removing all elements which are not equal to B .

Example 5.4 If $A \equiv [7, 7, [3, 5], [[4], 2]]$ and $B \equiv 7$ then $m_B(A) = 2$. Similarly if $C \equiv [3, 5]$ then $m_C(A) = 1$. But if $D \equiv [8]$ then $m_D(A) = 0$.

By recording multiplicities we may often express a box more simply; we do that with a *left subscript* denoting the multiplicity. We can rewrite the box A in the example above as $A = [2]7, [1][3, 5], [1][4], [2]]$, or simply $A = [2]7, [3, 5], [[4], 2]]$. The statement that 7 has multiplicity 2 in A can be rephrased as: the box 7 **occurs two times** in the box A .

Proposition 6 If A and B are boxes then $A \subseteq B$ precisely when each element X of C occurs with multiplicity less than or equal to the multiplicity of that element X in D .

Example 5.5 The box $[0, 5, 5, 17] = [0, 2]5, [1]17$ is a subbox of $[0, 0, 3, 5, 5, 17] = [2]0, [3, 2]5, [1]17$, so that $[0, 5, 5, 17] \subseteq [0, 0, 3, 5, 5, 17]$.

If a box C occurs as an element of A with multiplicity n , and occurs as an element of B with multiplicity m , then it will occur in $A + B$ with a multiplicity of $m + n$ times. So we can then perform addition of general boxes in terms of addition of (natural number) multiplicities.

Example 5.6 So for example if $X \equiv [8]0, [3]2, [2][[4]]]$ and $Y \equiv [3]0, [7]1, [5]2, [1][[4]]], [9][8]]$ then $X + Y = [11]0, [7]1, [8]2, [3][[4]]], [9][8]]$.

5.4 Multiples, linear combinations and spans

Natural numbers serve as scalar multipliers for any box A . We define **multiples** inductively: $0A \equiv 0$ and $(n + 1)A \equiv nA + A$ for any natural number n so that $1A \equiv A$ and $2A \equiv A + A$ and so on. Explicitly, nA is the box obtained by adding n copies of A together. The following multiple laws then follow directly from the commutativity and associativity of addition:

$$mA + nA = (m + n)A \quad \text{and} \quad m(A + B) = mA + mB$$

for any natural numbers m and n and boxes A and B . In both expressions, multiplication of a box by a scalar takes precedence over addition of boxes.

The notion of (natural number) multiple allows us to introduce a basic idea from linear algebra into this framework.

Definition A box M is a **linear combination of boxes** $A, B, \dots, K$ precisely when

$$M = aA + bB + \dots + kK$$

for some natural numbers $a, b, \dots, k$. Equivalently we declare that M **belongs to the span of** $X \equiv [A, B, \dots, K]$, and write $M \in \text{span}(X)$.

The elements of a box X are themselves in $\text{span}(X)$.

Example 5.7 The box $M \equiv [0, 5, 5, 17]$ is in $\text{span}(X)$ where $X = [[0], [5], [17]]$ as $M = [0] + 2[5] + [17]$. We also note that the multiple of 1 is not necessarily included explicitly.

Linear combination of boxes which belong to $\text{span}(X)$ for some box X , also belongs to $\text{span}(X)$. Determining whether a box Y belongs to $\text{span}(X)$ is in general a non-trivial computational problem — the box-arithmetic analog of solving a linear system over $\mathbb{N}$, which is harder than over a field since subtraction is unavailable. We leave explicit algorithms for a subsequent treatment.

An important variant of this is the following.

Definition A box M is a **ratio-linear combination** (or **r-linear combination**) of boxes $A, B, \dots, K$ precisely when there exist natural numbers $a, b, \dots, k$ and m such that

$$mM = aA + bB + \dots + kK.$$

This is equivalent to saying that M lies in the non-negative rational span of $\{A, B, \dots, K\}$ — that is, M can be expressed as a non-negative rational linear combination of $A, B, \dots, K$ — but expressed here entirely over the natural numbers by clearing denominators. Equivalently we declare that M **belongs to the r-span of** $X \equiv [A, B, \dots, K]$, written $M \in r\text{-span}(X)$. This notion is closed under taking further r-linear combinations, since clearing denominators again reduces any such combination to the required form.

Sets

In standard mathematics, multisets are typically defined as secondary objects — functions from a set to $\mathbb{N}$ recording multiplicities. In Box Arithmetic the hierarchy is reversed: sets are special cases of boxes, namely those boxes in which every element has multiplicity exactly 1. This reversal is not merely notational; it reflects the view that multiplicity is the more primitive notion.

For us sets are secondary to multisets/boxes, but they are still very important.

Definition A **set** is a box s in which each element $A \in s$ has multiplicity 1. If a box $s = [A, B, \dots, K]$ is a set, then we will write also $s = \{A, B, \dots, K\}$ using brace brackets. If a set has k elements, then we say it is a **k -set**.

We reserve brace brackets $\{ \}$ exclusively for sets, just as we reserve box brackets $[]$ exclusively for boxes, so the notation itself signals whether multiplicities are present.

Example 6.1 The box $A = [7, 2, [3, 3], 0]$ is a 4-set, and we may write $A = \{7, 2, [3, 3], 0\}$.

A subbox of a set S is called a **subset of S** . Sets are naturally partially ordered by inclusion.

For any natural number n , we define the **standard interval set** $\{0..n\} \equiv \{0, 1, 2, \dots, n\}$. More generally for natural numbers k and n with $k \leq n$ we define the **interval set**

$$\{k..n\} \equiv \{k, k+1, k+2, \dots, n\}.$$

Definition For any box M , the **support set of M** , denoted $\text{set}(M)$, is the set consisting of the elements of M , but now all with multiplicity 1.

Proposition 7 (Distributive laws) For any boxes A, B and C we have

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad \text{and} \quad A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

and

$$A + (B \cup C) = (A + B) \cup (A + C) \quad \text{and} \quad A + (B \cap C) = (A + B) \cap (A + C).$$

These laws just reduce to corresponding relations for natural numbers involving the operations $+$, $\min$ and $\max$.

Proof. The first two identities are the standard distributive laws of a lattice, and follow from the distributivity of $\max$ over $\min$ and $\min$ over $\max$ (Proposition 18). The second two — connecting addition with union and intersection — are distinctive to the multiset setting and follow from the identities

$$\max(a, b) + \max(a, c) = a + \max(b, c) \quad \text{and} \quad \min(a, b) + \min(b, c) = a + \min(b, c).$$

6.3 The lattice of subsets

We may also consider for a set S , the set $P[S]$ consisting of all of its subsets. This power set has the structure of a lattice.

6.4 Inclusion/Exclusion for boxes

There is also a wide ranging extension of the Inclusion/Exclusion Principle here.

Proposition 8 (Inclusion/Exclusion) For any boxes A and B we have

$$(A \cup B) + (A \cap B) = A + B$$

and for any boxes A, B and C we have

$$(A \cup B \cup C) + (A \cap B) + (A \cap C) + (B \cap C) = A + B + C + (A \cap B \cap C)$$

and there are natural extensions for more boxes.

Proof. The two-variable identity follows from the identity

$$\max \blacksquare(a, b) + \min(a, b) = a + b$$

applied elementwise. The three-variable identity follows similarly, and the pattern extends to any finite collection of boxes.

These Inclusion/Exclusion identities hold at the level of boxes themselves, not merely for their sizes. Taking sizes on both sides recovers the classical inclusion-exclusion principle for counts. The box-level versions are strictly stronger: they carry full multiplicity information rather than just totals.

Example 6.3 For boxes A and B we immediately deduce that $s[A \cup B] + s[A \cap B] = s[A] + s[B]$.

Example 6.4 For boxes A, B and C we immediately deduce that $s[A \cup B \cup C] + s[A \cap B] + s[A \cap C] + s[B \cap C] = s[A] + s[B] + s[C] + s[A \cap B \cap C]$.

Multiplication and higher operations

Multiplication is the natural successor to the addition operation, and holds for all boxes. Just like addition, it is not intrinsically a binary operation.

Definition If $A, B, \dots, K$ are boxes, then the **product** $A \times B \times \dots \times K$ is the box consisting of all possible sums of the elements of the boxes $A, B, \dots, K$, that is

$$A \times B \times \dots \times K \equiv [a + b + \dots + k : a \in A, b \in B, \dots, k \in K].$$

In case one or more of $A, B, \dots, K$ equals 0, the product has no elements in it, so is 0.

The meaning of this notation is the same as in classical set theory, where the colon ":" is to be read as "such that" and what precedes the colon is an object which is going to be an element of the box, and what follows it is a condition or conditions on that object. However as opposed to classical set theory, here multiplicities are maintained, so that we go through all possibilities without removing duplicates.

We note that multiplication is not intrinsically a binary operation.

Example 7.1 If $M = [a, a, K]$ and $N = [a, b]$ and $P = [K, L]$ with a, K, b and L boxes, then

$$M \times N \times P = \left[\begin{array}{l} a + a + K, a + a + L, a + b + K, a + b + L, a + a + K, a + a + L, a + b + K, a + b + L, \\ K + a + K, K + a + L, K + b + K, K + b + L \end{array} \right]$$

The product as defined is the **addition convolution** of the multisets: it records all ways of picking one element from each factor and summing them, with multiplicity.

Together with addition, multiplication gives the box world the structure of a *commutative semiring*: boxes under addition form a commutative monoid with identity 0, boxes under multiplication form a commutative monoid with identity $1 = [0]$, and multiplication distributes over addition. This is the box-arithmetic analog of a polynomial ring over $\mathbb{N}$, but defined for all boxes rather than just polynomials.

7.1 Properties of box multiplication

Proposition 9 (Product of sizes) If $A, B, \dots, K$ are boxes then $s[A \times B \times \dots \times K] = s[A] \times s[B] \times \dots \times s[K]$.

Proof. To find $s[A \times B \times \dots \times K]$ we replace each of its elements with 0. But each of these elements equals the sum $a + b + \dots + k$ for a particular choice of $a \in A, b \in B, \dots, k \in K$. Thus if we replace each element of $A, B, \dots, K$ by 0, the natural number $s[A \times B \times \dots \times K]$ does not change, but now the product $A \times B \times \dots \times K$ becomes the product $s[A] \times s[B] \times \dots \times s[K]$. This gives the identity.

Proposition 10 (Cancellation property) if A, B and C are boxes, with $A \neq 0$, and $A \times B = A \times C$, then $B = C$.

Proposition 11 (Zero product property) If $A, B, \dots, K$ are boxes and $A \times B \times \dots \times K = 0$, then we must have at least one of $A, B, \dots, K$ equal to 0.

Proposition 12 (One product property) If $A, B, \dots, K$ are boxes and $A \times B \times \dots \times K = 1$, then we must have $A = B = \dots = K = 1$.

Proof. Since $s[1] = 1$ has exactly one element, the Product of sizes proposition implies that $s[A] = s[B] = \dots = s[K] = 1$ and if a product of natural numbers is equal to 1 then each of them must be 1.

Proposition 13 The product of natural numbers is a natural number.

Proof. This follows since the elements of natural numbers are all 0, and $0 + 0 = 0$.

Multiplication of natural numbers (as boxes) then becomes the usual familiar operation, for example $3 \times 4 = [0, 0, 0] \times [0, 0, 0, 0] = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0] = 12$.

The previous notion of a multiple of a box may now be reinterpreted as a multiplication by a natural number: for a box A , we now have $2A \equiv A + A = 2 \times A$ and $3A \equiv A + A + A = 3 \times A$, and in general for any natural number k we have $kA \equiv A + A + \cdots + A = k \times A$ with k terms. For consistency we note that also $0A \equiv 0 \times A = 0$ for any box A . We may also augment the multiple laws to include the rule that for natural numbers m and n and a box A we have

$$n(mA) = (m \times n)A.$$

7.2 Powers of boxes

Similarly we now may introduce **powers** of a box.

Definition For a box A , we define $A^2 \equiv A \times A$ and $A^3 \equiv A \times A \times A$, and in general by induction $A^{k+1} \equiv A^k \times A$ so that $A^n = A \times A \times \cdots \times A$ with n terms. For consistency we define $A^0 \equiv 1$ for any box A . This means in particular that we also agree that $0^0 = 1$.

We then have the **power laws**.

Proposition 14 For any natural numbers k and m and any boxes A and B

$$A^k \times A^m = A^{k+m} \quad A^k \times B^k = (A \times B)^k \quad \text{and} \quad (A^k)^m = A^{k \times m}.$$

More generally if $A, B, \dots, K$ are boxes, then

$$(A \times B \times \cdots \times K)^n = A^n \times B^n \times \cdots \times K^n.$$

Example 7.2 If $M = [A, B]$ then

$M^2 = [A + A, A + B, B + A, B + B] = [2A, {}_2(A + B), 2B]$ while $M^3 = [A + A + A, A + A + B, A + B + A, A + B + B, B + A + A, B + A + B, B + B + A, B + B + B] = [3A, {}_3(2A + B), {}_3(A + 2B), 3B]$.

Example 7.3 If $M = [A, B, C]$ then

$$M^2 = [A + A, A + B, A + C, B + A, B + B, B + C, C + A, C + B, C + C] = [2A, 2B, 2C, {}_2(A + B), {}_2(A + C), {}_2(B + C)].$$

We may also multiply boxes by working with multiplicities, such as

$$[{}_n A, {}_m B] \times [{}_p C, {}_q D] = [{}_{n \times p} (A + C), {}_{n \times q} (A + D), {}_{m \times p} (B + C), {}_{m \times q} (B + D)].$$

Here both addition and multiplication of natural numbers are generally required.

There is a natural *hierarchy of arithmetical operations* on boxes, starting with addition $\times_0 \equiv +$ and then multiplication $\times_1 \equiv \times$, with each generating the next by the same uniform schema: the *successor* of the operation $\times_n$ is the operation $\times_{n+1}$ defined by collecting all possible $\times_n$ -combinations of elements:

$$A \times_{n+1} B \times_{n+1} \cdots \times_{n+1} K \equiv [a \times_n b \times_n \cdots \times_n k : a \in A, b \in B, \dots, k \in K].$$

Addition generates multiplication, multiplication generates the caret operation $\times_2 \equiv \wedge$ (Section 8), and so on. Every operation $\times_n$ in this hierarchy is commutative, associative, and distributes over addition $\times_0$ — for the same reason at each level: elements are collected without regard to order. This contrasts with the classical hyperoperation hierarchy (exponentiation, tetration, pentation, ...) which is neither commutative nor acts uniformly on all boxes. The higher operations in the box hierarchy lead to genuinely unfamiliar algebra whose full theory remains largely unexplored.

7.3 The two term domination Lemma

We introduce a fundamental identity for natural numbers, which is a special case of the AM-GM inequality. It is usually proven in the context of integer arithmetic involving subtraction, but it holds in this more elementary situation just involving natural numbers.

Lemma 7.1 (Two term domination) For any natural numbers m and n we have

$$2(m \times n) \leq m^2 + n^2.$$

Proof. If m and n are natural numbers, then either $m \leq n$ or $n \leq m$ (or both). Without loss of generality, suppose that $m \leq n$ so that $m + d = n$ for some natural number d . Then $2(m \times n) = 2 \times m \times (m + d) = 2m^2 + 2(m \times d)$ while $m^2 + n^2 = m^2 + (m + d)^2 = 2m^2 + 2(m \times d) + d^2 = 2(m \times n) + d^2$. Hence $2(m \times n) \leq m^2 + n^2$.

Note that the result does not hold for boxes more generally, since arbitrary boxes are not necessarily comparable, which is what we required for the natural number case.

Corollary 2 Suppose that a, b, c and d are natural numbers. Then

$$(a \times c + b \times d)^2 \leq (a^2 + b^2) \times (c^2 + d^2).$$

Proof. If we expand the left hand side we get

$(a^2 \times c^2) + (b^2 \times d^2) + 2(a \times c \times b \times d)$. If we expand the right hand side we get $(a^2 \times c^2) + (b^2 \times d^2) + (a^2 \times d^2) + (b^2 \times c^2)$. So the inequality reduces to

$$2(a \times c \times b \times d) \leq (a^2 \times d^2) + (b^2 \times c^2)$$

But this is just the two term domination Lemma 2 $2(m \times n) \leq m^2 + n^2$ with $m = a \times d$ and $n = b \times c$.

7.4 Power linear and ideal linear combinations

Definition If X is a box, then any box of the form

$A = a_0 + a_1X + a_2X^2 + \cdots + a_nX^n$ for natural numbers $a_0, a_1, \dots, a_n$, will be called a **power linear combination** of X .

Then any sum or product of such power linear combinations of X will also be a power linear combination of X . More generally, we can define the box A to be a **power linear combination** of boxes $X, Y, \dots, W$ to mean that A is a linear combination of products of powers of $X, Y, \dots, W$.

We can extend this further.

Definition If $X, Y, \dots, W$ are boxes, then any box of the form

$A = x \times X + y \times Y + \cdots + w \times W$ for some boxes $x, y, \dots, w$ will be called an **ideal linear combination** of $X, Y, \dots, W$.

This generalises Definition ?? by allowing the coefficients to be arbitrary boxes rather than natural numbers. The name reflects the analogy with ring theory: in a commutative ring, the ideal generated by elements $X, Y, \dots, W$ consists precisely of all such linear combinations with ring-valued coefficients. Here the box semiring plays the role of the ring. The ideal linear combinations of $X, Y, \dots, W$ are closed under addition and under power linear combinations — making them the box-arithmetic analog of an ideal. These notions are introduced as a foundation for the polynomial algebra which we now introduce.

Polynumber algebra

We now introduce polynomial arithmetic in a new way: polynomials become concrete finite objects — boxes of natural numbers — rather than formal expressions involving an abstract variable or functions on a domain. This gives polynomials decidable equality and explicit finite representations, resolving the usual ambiguity about whether two polynomial expressions are "the same."

Definition A **polynumber** is a box of natural numbers. If p is a non-zero polynumber then its **degree** $\deg[p]$ is the largest natural number that appears as an element of it. The degree of the zero polynumber 0 is left undefined. Note that $\deg[1] = \deg[[0]] = 0$.

Example 8.1 The boxes $p \equiv [0, 2]$, $r \equiv [3, 5, 3]$ and $s \equiv [0, 0, 6, 0, 1, 3, 3]$ are polynumbers, with $\deg(p) = 2$, $\deg(r) = 5$ and $\deg(s) = 6$.

Addition of polynumbers is just addition of boxes, so that for example with p and r as above, $p + r = [0, 2, 3, 3, 5]$. **Multiplication of polynumbers** is just multiplication of boxes, so that for example $p \times r = [0, 2] \times [3, 5, 3] = [0 + 3, 0 + 5, 0 + 3, 2 + 3, 2 + 5, 2 + 3] = [3, 5, 3, 5, 7, 5]$.

8.1 The polynumber alpha

We now introduce the simplest polynumber which is not actually a natural number.

Definition The polynumber **alpha** $\alpha \equiv [1]$ is a specific concrete box, not an abstract variable or formal symbol.

Proposition 15 For any natural number n we have $\alpha^n = [n]$.

Proof. By induction: $\alpha^0 = 1$ by convention and $\alpha^1 = [1]$ by definition. If $\alpha^n = [n]$ then $\alpha^{n+1} = \alpha^n \times \alpha = [n] \times [1] = [n+1]$.

Proposition 16 Every nonzero polynumber p has a unique representation

$$p = p_0 + p_1\alpha + \cdots + p_n\alpha^n$$

where $n = \deg [p]$ and the coefficients $p_0, p_1, \dots, p_n$ are natural numbers with $p_n \neq 0$.

Proof. The elements of p are natural numbers with various multiplicities. Setting p_k to be the multiplicity of k in for $0 \leq k \leq n = \deg [p]$ we may write $p = p_0 + p_1 [1] + \cdots + p_n [n]$ and using $\alpha^k = [k]$ gives the result.

We say that such an expression is in **standard polynumber form** if the powers of α are increasing as we read from left to right: this will be our usual convention. Addition and multiplication of polynumbers then agree with the familiar operations on polynomials with natural number coefficients. The algebra of polynumbers is thus the same as the polynomial algebra $\mathbb{N}[\alpha]$ but with α now a concrete object rather than a formal symbol.

Example 8.2 With p and r as in the previous Example, we have
 $p + r = (1 + \alpha^2) + (2\alpha^3 + \alpha^5) = 1 + \alpha^2 + 2\alpha^3 + \alpha^5$ and
 $p \times r = (1 + \alpha^2) \times (2\alpha^3 + \alpha^5) = [3, 3, 5, 5, 5, 7] = 2\alpha^3 + 3\alpha^5 + \alpha^7$.

8.2 Binomial coefficients

The Binomial theorem is accessible in terms of writing expansions for $(1 + \alpha)^n$ for a natural number n .

Definition The **binomial coefficients** $\binom{n}{k}$ for natural numbers n and k are defined to be the coefficients in the expansion

$$(1 + \alpha)^n \equiv \binom{n}{0} + \binom{n}{1}\alpha + \binom{n}{2}\alpha^2 + \cdots + \binom{n}{n}\alpha^n.$$

No factorials, quotients, or prior theory of combinations is required: the coefficients are

simply read off from a concrete polynumber computation. We set $\binom{n}{k} = 0$ if $k > n$ consistent with the degree of $(1 + \alpha)^n$ being n .

The **Pascal recurrence**

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$$

follows by expanding $(1 + \alpha)^{n+1} = (1 + \alpha)^n \times (1 + \alpha)$ and comparing coefficients of α^{k+1} on both sides. The boundary values $\binom{n}{0} = \binom{n}{n} = 1$ follow from the fact that $(1 + \alpha)^n$ has constant term 1 and leading coefficient 1. The symmetry $\binom{n}{k} = \binom{n}{l}$ if $k + l = n$ follows from the boundary conditions and the Pascal recurrence.

8.3 Evaluation and composition

Definition The **evaluation** of the polynumber $p = p_0 + p_1\alpha + p_2\alpha^2 + \dots + p_n\alpha^n$ on the box A is the box

$$p(A) \equiv p_0 + p_1A + p_2A^2 + \dots + p_nA^n.$$

For polynumbers p and q , a natural number λ and a box A , we have $(p + q)(A) = p(A) + q(A)$ and $(\lambda p)(A) = \lambda p(A)$ and $(p \times q)(A) = p(A) \times q(A)$. In particular if we evaluate a polynumber p on a natural number n , then the result $p(n)$ is also a natural number, and the above relations are satisfied. There is another important special case.

Definition If $p = p_0\alpha^0 + p_1\alpha^1 + p_2\alpha^2 + \dots + p_n\alpha^n$ is a polynumber and q is another polynumber, then the **composition** $p \circ q$ is the polynumber

$$p \circ q \equiv p(q) = p_0 + p_1q + p_2q^2 + \dots + p_nq^n.$$

The algebra of polynumbers is richly expanded by considering also this composition operation together with addition and multiplication. While composition is not commutative, it is associative, in the following sense: if p, q and r are polynumbers then

$$p \circ (q \circ r) = (p \circ q) \circ r.$$

Composition distributes over addition and scalar multiplication from the right: for any polynumbers p, q, r and natural number λ

$$(p + q) \circ r = (p \circ r) + (q \circ r) \quad \text{and} \quad (\lambda \times p) \circ r = \lambda \times (p \circ r).$$

Left distributivity fails in general: $p \circ (q + r) \neq p \circ q + p \circ r$. For example with $p = \alpha^2$, $q = \alpha$ and $r = 1$, $p \circ (q + r) = (1 + \alpha)^2 = 1 + 2\alpha + \alpha^2$ while $p \circ q + p \circ r = \alpha^2 + 1$. This asymmetry is a fundamental feature of composition and reflects the non-linearity of polynomial substitution.

8.4 More general size recipes

The size recipe $s[A]$ replaces each element of A with the empty box 0, discarding one level of internal structure. This generalises naturally to a hierarchy of recipes s_n , one for each natural number n , each discarding structure at a different depth. The clearest picture comes from the tree representation of boxes: s_n **prunes** the rooted tree at depth n , removing all nodes below level n and replacing them with leaves. So s_0 collapses everything to 0, $s_1 = s$ records only the number of elements, s_2 records the sizes of the elements, and so on.

Definition For a box A , define the recipe s_0 on boxes by the rule that $s_0[A] \equiv 0$, so that given a box A we replace A with 0, or equivalently remove all elements of A . For a natural number n define the recipe s_{n+1} on boxes by the rule that $s_{n+1}[A]$ is the box where each element X of A is replaced with $s_n[X]$. In particular $s_1[A]$ agrees with the size $s[A]$ defined above. The value of $s_2[A]$ is obtained by replacing each element X of A with $s_1[X]$, which is the size of X .

Note that s_n is defined by recursion on the index n with s_0 as the base case and s_{n+1} expressed in terms of s_n . This is a different pattern from structural recursion on boxes: here the recursion runs on a natural number parameter, with each s_n itself a non-recursive recipe on boxes.

Example 8.3 We have $s_2[3] = s_2[[0, 0, 0]] = [0, 0, 0] = 3$ since the size $s[0]$ of the empty box is 0. Also $s_2[[0, 0, 2, 5, 5, 5]] = [0, 0, 2, 5, 5, 5]$ since the size of a natural number n is $s[n] = n$. However $s_2[[[0, 0], [0, 2, 3, 3], [5]]] = s_2[\alpha_0^2 + \alpha_0\beta_2\alpha_3^2 + \alpha_5] = [2, 4, 1] = \alpha + \alpha^2 + \alpha^4$.

Proposition 17 The recipe s_2 agrees with s_1 on polynumbers: $s_2[p] = s_1[p]$ for any polynumber p

Proof. A polynumber is a box of natural numbers. For any natural number n , $s_1[n] = n$ since $n = [0, \dots, 0]$ has n elements each of which is 0, so replacing each element with 0 gives n again. Hence $s_2[p] = [s_1(e) : e \in p] = [e : e \in p] = p$.

More generally s_{n+1} agrees with s_n on boxes of level at most n in the hierarchy — natural numbers at level 1, polynumbers at level 2, multinumbers at level 3, and so on.

The recipe s_n is thus the identity of the first n levels and projects higher-level boxes onto level n .

8.5 Summations of boxes and the derivative

The operation of addition can also be introduced just for boxes, with regard to their elements.

Definition The **summation recipe** Σ takes boxes to boxes, and is defined as follows: if $M = [A, B, \dots, K]$ is a non-zero box, then the **summation of** M is

$$\Sigma[M] \equiv A + B + \dots + K$$

and also we define $\Sigma(0) \equiv 0$.

We will also use ordinary summation notation to write

$$\Sigma[M] \equiv \sum_{m \in M} m.$$

Of course we need to observe that this is well-defined, in other words independent of the order of the elements.

Example 8.4 If $M = [\square, \square, \square, \square, [[\square, \square]], [[\square, \square, \square]]] = [0, 0, 0, 0, 2, [3]]$ then

$$\begin{aligned} \Sigma[M] &= \square + \square + \square + \square + [\square, \square] + [[\square, \square, \square]] = 0 + 0 + 0 + 0 + 2 + [3] \\ &= [\square, \square, [\square, \square, \square]] = [0, 0, 3]. \end{aligned}$$

Proposition 18 For boxes M and N and a natural number λ we have $\Sigma[M + N] = \Sigma[M] + \Sigma[N]$ and $\Sigma(\lambda \times M) = \lambda \times \Sigma(M)$.

Proof. If $M = [A, B, \dots, K]$ and $N = [P, Q, \dots, W]$ then

$M + N = [A, B, \dots, K, P, Q, \dots, W]$ and

$\Sigma[M + N] = A + B + \dots + K + P + Q + \dots + W = \Sigma[M] + \Sigma[N]$. The relation

$\Sigma(\lambda \times M) = \lambda \times \Sigma(M)$ is established by repeated use of the first relation (induction).

This says that Σ is a **linear** recipe: it respects addition. Linearity will be a recurring and important property of recipes throughout the paper. The effect of summation on a product is more complicated.

Proposition 19 (Product rule for Σ) For boxes M and N we have

$$\Sigma(M \times N) = s[M] \times \Sigma(N) + s[N] \times \Sigma(M).$$

Proof. An element of $M \times N$ has the form $a + b$ with $a \in M, b \in B$. Summing over all such pairs

$$\Sigma(M + N) = \sum_{a \in M, b \in N} (a + b) = s[N] \times \Sigma(M) + s[M] \times \Sigma(N)$$

since each a appears $s[N]$ times and each b appears $s[M]$ times.

Proposition 20 (Power rule for Σ) For a box M and a natural number λ we have

$$\Sigma(\lambda \times M) = \lambda^{s[M]} \times \Sigma(M).$$

Proof. Apply the previous Proposition and use Induction.

These are the box arithmetic analogues of the Leibniz product and power rules from calculus, and we can make a more direct connection by looking at the case when the boxes are polynumbers.

Example 8.5 If p is a polynumber, say

$p = p_0 + p_1\alpha + \cdots + p_k\alpha^k = \lfloor_{p_0} 0, p_1 \ 1, \cdots, p_k \ k \rfloor$ then $\Sigma(p)$ is a natural number given by

$$\Sigma(p) = p_0 \times 0 + p_1 \times 1 + p_2 \times 2 + \cdots + p_k \times k.$$

In particular if m is a natural number, then $\Sigma(m) = 0$.

This motivates us to introduce the following notion.

Definition The **derivative recipe** D takes polynumbers to polynumbers and is defined as follows. Suppose that $p = p_0 + p_1\alpha + \cdots + p_{k+1}\alpha^{k+1}$ for some natural number k , then

$$D[p] \equiv p_1 + 2p_2 \times \alpha + \cdots + (k+1)p_{k+1} \times \alpha^k$$

and also define $D[0] \equiv 0$.

One can check that then for polynumbers p and q , and a natural number λ , we have $D[p + q] = D[p] + D[q]$ and $D[\lambda \times p] = \lambda \times D[p]$.

Example 8.6 If $p = [0, 0, 0, 4, 7, 7] = [{}_3 0, {}_1 4, {}_2 7] = 3 + \alpha^4 + 2\alpha^7$ then $D[p] = 4\alpha^3 + 7\alpha^6$

Proposition 21 For polynumbers p and q we have

$$D[p \times q] = p \times D[q] + q \times D[p].$$

Proof. Since $p \times q$ is a polynumber and D is linear, it suffices to show this when $p = \alpha^m$ and $q = \alpha^n$. If one of m and n are 0 then the formula is immediate. Otherwise write $m = k + 1$ and $n = l + 1$ for some natural numbers k and l , so that

$$D[p \times q] = D[\alpha^{k+1} \times \alpha^{l+1}] = D[\alpha^{k+l+2}] = (k + l + 2) \times \alpha^{k+l+1}$$

while

$$p \times D[q] + q \times D[p] = \alpha^{k+1} \times (l + 1) \alpha^l + \alpha^l \times (k + 1) \alpha^k = (k + l + 2) \times \alpha^{k+l+1}$$

also.

We can now restate the formula for the summation of a polynumber p as

$$\Sigma(p) = D[p](1)$$

and when we combine that with the product rule we get

$$\Sigma(p \times q) = |q| \times \Sigma(p) + |p| \times \Sigma(q).$$

Example 8.7 If $p = [3, 3, 4]$ and $q = [1, 2]$ we have $p \times q = [3 + 1, 3 + 2, 3 + 1, 3 + 2, 4 + 1, 4 + 2] = [4, 5, 4, 5, 5, 6]$ so that $\Sigma(p \times q) = 29$ while $s(q) \times \Sigma(p) + s(p) \times \Sigma(q) = 2 \times 10 + 3 \times 3 = 29$.

Example 8.8 If $p = [A, B]$ and $q = [C, D, E]$ then $\Sigma(p \times q) = \Sigma([A + C, A + D, A + E, B + C, B + D, B + E]) = A + C + A + D + A + E + B + C + B + D + B + E = 3A + 3B + 2C + 2D + 2E$

8.6 Truncations of polynumbers

Definition For a natural number k we define the **k -th degree truncation recipe** taking polynumbers to polynumbers as follows: for a polynumber p , the polynumber $T^k[p]$ is obtained from p by removing all elements which are strictly greater than k . This is a recipe which is **linear** in the sense that $T^k[p + q] = T^k[p] + T^k[q]$ and $T^k[\lambda \times p] = \lambda \times T^k[p]$ for polynumbers p and q and a number λ .

Proposition 22 (Truncation formula) For polynumbers p and q and a natural number k we have

$$T^k[p \times q] = T^k(T^k[p] \times T^k[q]).$$

This is encoding the fact that the truncation of a product $p \times q$ up to degree k only depends on the truncations of p and q up to degree k .

Proof. An element of $p \times q$ has the form $a + b$ with $a \in p$ and $b \in q$. Now we have $a + b \leq k$ implies that both $a \leq k$ and $b \leq k$ so that the elements of $p \times q$ that survive truncation at k are exactly those $a + b$ with $a \in T^k[p]$ and $b \in T^k[q]$ satisfying $a + b \leq k$, which are precisely the elements of $T^k(T^k[p] \times T^k[q])$.

Note that T^k is linear but not multiplicative: $T^k[p \times q] \neq T^k[p] \times T^k[q]$ in general, since the right side may contain terms of degree greater than k . The truncation captures exactly this failure.

Definition For a natural number n we let $\mathcal{P}_n$ be the **degree n truncation algebra** consisting of polynumbers of degree up to n , with the usual addition, and product $\times_n$ defined by

$$p \times_n q \equiv T_n(p \times q).$$

Example 8.9 If $n = 4$ then $p = 2 + \alpha - \alpha^4$ and $q = \alpha^2 + 7\alpha^4$ are in $\mathcal{P}_4$. Then in $\mathcal{P}_4$ we have $p + q = 2 + \alpha + \alpha^2 + 6\alpha^4$ while $p \times_4 q = (2 + \alpha - \alpha^4) \times_4 (\alpha^2 + 7\alpha^4) = T_4(2\alpha^2 + \alpha^3 + 14\alpha^4 + 7\alpha^5 - \alpha^6 - 7\alpha^8) = 2\alpha^2 + \alpha^3 + 14\alpha^4$.

There is a useful variant of equality for polynumbers that depends on a natural number n and uses the above truncation.

Definition For polynumbers p and q and a natural number n , we define the **degree n equality** $p =_n q$ precisely when $T^n(p) = T^n(q)$.

Then we can rewrite the truncation formula above as

$$p \times q =_k T^k[p] \times T^k[q].$$

The truncation algebras $\mathcal{P}_n$ allow us the possibility of a genuinely finite substitute for the classical theory of "formal power series". A "formal power series" $\sum_{k=0}^{\infty} a_k \alpha^k$ is an inherently infinite object — it cannot be written down in full, only described by a rule. The truncation algebra $\mathcal{P}_n$ captures all information about polynomial arithmetic up to degree n in a completely finite object with decidable equality.

In classical algebra, $\mathcal{P}_n$ is typically constructed as the quotient of an infinite-dimensional polynomial algebra by the ideal of polynomials vanishing to order $n + 1$ — a construction requiring both an infinite algebra and an infinite ideal. Here $\mathcal{P}_n$ is defined directly and finitely, with no infinite objects in sight. The degree n equality $p =_n q$ then gives a precise finite substitute for the notion of two power series "agreeing to order n ", which is the foundational idea behind asymptotic analysis and the calculus of infinitesimals, here placed on an entirely finite footing.

Multinumbers

Definition A **multinumber** is a box of polynumbers.

Example 9.1 The multinumber $A = [0, 4, 4, [3, 5], [0, 7, 7, 6]]$ has the 5 elements $0, 4, 4, [3, 5]$ and $[0, 7, 7, 6]$, each of which is a polynumber, specifically $[3, 5] = \alpha^3 + \alpha^5$ and $[0, 7, 7, 6] = 1 + \alpha^6 + 2\alpha^7$. Thus we could rewrite $A = [0, 4, 4, \alpha^3 + \alpha^5, 1 + \alpha^6 + 2\alpha^7]$.

Since natural numbers are polynumbers, we have the containment hierarchy: natural numbers $\subset$ polynumbers $\subset$ multinumbers, corresponding to levels 1, 2 and 3 in the box hierarchy. Multinumbers are closed under addition and multiplication — since polynumbers are closed under addition, the product of two multinumbers, formed by collecting all pairwise sums of elements, is again a multinumber. As we shall see, multinumbers encode the algebra of polynomials in finitely many variables with natural number coefficients, but with the variables themselves being concrete boxes rather than formal symbols.

Example 9.2 If $B = [2, 1 + \alpha]$ and $C = [5, \alpha^2, 1 + \alpha]$ then $B + C = [2, 1 + \alpha, 5, \alpha^2, 1 + \alpha]$ while $B \times C = [7, 3 + \alpha^2, 3 + \alpha, 6 + \alpha, 1 + \alpha + \alpha^2, 2 + 2\alpha]$.

The **base multinumbers** are defined for each natural number n by $\beta_n \equiv [\alpha^n] = \llbracket n \rrbracket$. In particular $\beta_0 \equiv \alpha = [1]$ reinterpreting the polynumber α as a base multinumber. These base multinumbers play the role of independent variables in multinomial algebra: any multinumber can be expressed as a finite sum of monomials $c\beta_{n_1}^{e_1}\beta_{n_2}^{e_2}\cdots\beta_{n_r}^{e_r}$ with natural number coefficients c , involving only finitely many base multinumbers.

Example 9.3 The multinumber A from the previous Example may be written as

$$A = [0, 4, 4, [3, 5], [0, 7, 7, 6]] = [0] + [4] + [4] + [[3]] \times [[5]] + [[0]] \times [[7]] \times [[7]] \times [[6]] \\ = 1 + 2\beta_0^4 + \beta_3\beta_5 + \beta_0\beta_6\beta_7^2.$$

We see that the polynumber $[3, 5] = \alpha^3 + \alpha^5$ corresponds to the monomial term $\beta_3\beta_5$ while the polynumber $[0, 7, 7, 6] = 1 + \alpha^6 + 2\alpha^7$ corresponds to the monomial term $\beta_0\beta_6\beta_7^2$.

For the polynumber $p \equiv k\beta^m$ for natural numbers k and m , the corresponding multinumber is

$$[p] = [[m] + [m] + \cdots + [m]] = [[m]] \times [[m]] \times \cdots \times [[m]] = \beta_m^k$$

with k copies in both the sum and the product. It follows that if q is the polynumber $q \equiv q_0 + q_1\alpha + q_2\alpha^2 + \cdots + q_n\alpha^n$ for some natural number n and natural numbers $q_0, q_1, \dots, q_n$, then the multinumber $Q = [q]$ can be written as

$$Q = [q_0 + q_1\alpha + q_2\alpha^2 + \cdots + q_n\alpha^n] = [q_0] \times [q_1\alpha] \times \cdots \times [q_n\alpha^n] \\ = \beta_0^{q_0} \times \beta_1^{q_1} \times \cdots \times \beta_n^{q_n} = \beta_0^{q_0} \beta_1^{q_1} \cdots \beta_n^{q_n}.$$

For a monomial $c\beta_{n_1}^{e_1}\beta_{n_2}^{e_2}\cdots\beta_{n_r}^{e_r}$ its degree is $e_1 + e_2 + \cdots + e_r$ (the total exponent) and its **extent** is $\max(n_1, n_2, \dots, n_r)$ (the highest variable index appearing). For a general multinumber m , the **degree of** m is the maximum degree among its monomials, and the **extent of** m is the maximum extent. Degree measures polynomial complexity; extent measures how far along the variable sequence the multinumber reaches.

Example 9.4 For the multinumber

$A = [0, 2, 4, [3, 5], [0, 7, 7, 6]] = 1 + 2\beta_0^4 + \beta_3\beta_5 + \beta_0\beta_6\beta_7^2$, the degree of A is the degree of the term $\beta_0\beta_6\beta_7^2$, which is 4, while the extent of A is the extent of that same term, which is 7.

9.1 Polynomial algebra in unbounded variables

Then multinumbers give us familiar polynomial algebra, based on addition and multiplication of the base multinumbers $\beta_0, \beta_1, \beta_2, \dots, \beta_n$ for any natural number n .

Example 9.5 If $B = [0, [3], [2, 4]] = 1 + \beta_3 + \beta_2\beta_4$ and $C = [[1, 1], [2, 4]] = \beta_1^2 + \beta_2\beta_4$ then

$$B + C = [0, [3], [2, 4]] + [[1, 1], [2, 4]] = [0, [3], [1, 1], [2, 4], [2, 4]] = 1 + \beta_1^2 + \beta_3 + 2\beta_2\beta_4$$

which agrees with $(1 + \beta_3 + \beta_2\beta_4) + (\beta_1^2 + \beta_2\beta_4) = 1 + \beta_1^2 + \beta_3 + 2\beta_2\beta_4$. Similarly

$$B \times C = [0, [3], [2, 4]] \times [[1, 1], [2, 4]] \\ = [0 + [1, 1], 0 + [2, 4], [3] + [1, 1], [3] + [2, 4], [2, 4] + [1, 1], [2, 4] + [2, 4]] \\ = [[1, 1], [2, 4], [3, 1, 1], [3, 2, 4], [2, 4, 1, 1], [2, 4, 2, 4]] \\ = \beta_1^2 + \beta_2\beta_4 + \beta_1^2\beta_3 + \beta_2\beta_3\beta_4 + \beta_1^2\beta_2\beta_4 + \beta_2^2\beta_4^2$$

which agrees with $(1 + \beta_3 + \beta_2\beta_4)(\beta_1^2 + \beta_2\beta_4) = \beta_1^2 + \beta_2^2\beta_4^2 + \beta_2\beta_4 + \beta_1^2\beta_3 + \beta_1^2\beta_2\beta_4 + \beta_2\beta_3\beta_4$.

9.2 Factors and divisibility

Definition If A, B and C are boxes with $A \times B = C$ then A and B are **factors** or **divisors** of C , and C is a **box multiple** of A and of B . In this case we write $A|C$ and $B|C$ and say that A and B **divide** C , or that C is **divisible by** A and B .

Example 9.6 Since $(1 + \alpha)(1 + \alpha^2) = 1 + \alpha + \alpha^2 + \alpha^3$ we have that $1 + \alpha$ and $1 + \alpha^2$ are factors of $1 + \alpha + \alpha^2 + \alpha^3$, while $1 + \alpha + \alpha^2 + \alpha^3$ is a box multiple of both $1 + \alpha$ and $1 + \alpha^2$.

Example 9.7 Since $0 \times A = 0$ for any box A , the box 0 is divisible by any box.

Divisibility is a partial order on boxes. Reflexivity is immediate: $A|A$ since $A \times 1 = A$. Transitivity is straightforward: if $A|B$ and $B|C$ then $A|C$ directly. Antisymmetry requires more care: if $A|B$ and $B|A$ then there exist boxes P and Q with $A \times P = B$ and $B \times Q = A$, giving $A \times P \times Q = Q$, hence $P \times Q = 1$ by cancellation, hence $P = Q = 1$ by Proposition 43, giving $A = B$.

For a box A , let $\text{div}(A)$ denote the set of divisors of A . Since $1 \times A = A$ for any box A , the box A always has factors 1 and A .

Definition A box A is **irreducible** precisely when it has exactly two factors, namely 1 and A . By this definition, 1 is not irreducible.

Definition A box A is **prime** precisely when it has the property that if $A|(B \times C)$ for boxes B and C , then either $A|B$ or $A|C$.

In classical ring theory, irreducibility and primality coincide in unique factorization domains — but the proof of this equivalence uses subtraction essentially, via Euclid's algorithm. Without subtraction, the two notions genuinely diverge: every prime box is irreducible, but irreducible boxes need not be prime. The next Example illustrates this concretely. The natural numbers are special: unique factorization holds, and irreducibility and primality coincide, giving the Fundamental Theorem of Arithmetic as a special case of box divisibility theory.

Example 9.8 The identity $1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 = (1 + \alpha)(1 + \alpha^2 + \alpha^4) = (1 + \alpha^3)(1 + \alpha + \alpha^2)$ has all four factors irreducible as boxes, but they are clearly not primes, because for example

$(1 + \alpha^3)$ divides the product $(1 + \alpha)(1 + \alpha^2 + \alpha^4)$ but does not divide either of the factors $(1 + \alpha)$ or $(1 + \alpha^2 + \alpha^4)$.

However for natural numbers, the notions of irreducible and prime coincide.

Definition A natural number which is prime is called a **prime number**.

Example 9.9 The prime numbers up to 100 are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89 and 97.

While unique factorization into primes or irreducibles does not hold for boxes in general, it does hold for natural numbers, and it is still of interest to inquire about factorizability in other circumstances.

Definition For a natural number n define the **primorial set**

$$P[n] \equiv \{p : p \text{ is a prime number and } p \leq n\}.$$

Its size $s(P[n])$ is defined to be $\pi[n]$.

Definition For a natural number n and a prime number p , define the **tower set**

$$T[p, n] \equiv \{p^m : m \text{ is a natural number and } p^m \leq n\}.$$

Note that $1 = p^0$ is an element of any such tower set.

These combinatorial sets organize the prime factorization data needed for the Fundamental Identity of Arithmetic in Section ??, where they appear as the index sets for caret products over primes and prime powers.

The caret operation

Definition The successor operation to multiplication $\times$ is what we call the **caret operation**, and denote by $\wedge$; for boxes $A, B, \dots, K$ we define

$$A \wedge B \wedge \dots \wedge K \equiv \lfloor a \times b \times \dots \times k : a \in A, b \in B, \dots, k \in K \rfloor$$

the box of all possible products between elements of A and elements of B .

The caret operation differs from classical exponentiation in three fundamental ways: it is commutative, so that $A \wedge B = B \wedge A$, and associative, so that $A \wedge (B \wedge C) = (A \wedge B) \wedge C$, and it distributes over addition for the same reason that multiplication does, so that $A \wedge (B + C) = (A \wedge B) + (A \wedge C) = A \wedge B + A \wedge C$. For any box A we have $0 \wedge A = 0$ since 0 has no elements, and $1 \wedge A = s(A)$ since multiplying any element of A by 0 gives 0, and there are $s(A)$ such elements.

When restricted to natural numbers, caret gives ordinary multiplication: $m \wedge n = \lfloor 0 \times 0 : 0 \in m, 0 \in n \rfloor = \lfloor 0 + 0 : 0 \in m, 0 \in n \rfloor = m \times n$. More generally we have the following.

Proposition 23 For any boxes A and B we have

$$s[A \wedge B] = s[A] \times s[B] = s[A] \wedge s[B].$$

If we consider the caret of polynumbers, we have $\lfloor m \rfloor \wedge \lfloor n \rfloor = \lfloor m \times n \rfloor$ which we can restate as $\alpha^m \wedge \alpha^n = \alpha^{m \times n}$ so this operation is distinct from multiplication on polynumbers.

Example 10.1 Here is a caret of more general polynumbers:

$$[1, 3, 2] \wedge [2, 4] = [1 \times 2, 1 \times 4, 3 \times 2, 3 \times 4, 2 \times 2, 2 \times 4] = [2, 4, 6, 12, 4, 8]$$

which in polynomial form is $(\alpha + \alpha^2 + \alpha^3) \wedge (\alpha^2 + \alpha^4) = (\alpha^2 + 2\alpha^4 + \alpha^6 + \alpha^8 + \alpha^{12})$.

This contrast is fundamental: polynumber multiplication adds exponents ($\alpha^m \times \alpha^n = \alpha^{m+n}$) while the caret multiplies them. This is analogous to the Dirichlet product in number theory, where multiplicative functions combine by multiplying their values at prime powers.

When we consider the caret operation acting on multinumbers, we discover something again a bit surprising: for natural numbers m and n we have

$$\beta_m \wedge \beta_n = \llbracket [m] \rrbracket \wedge \llbracket [n] \rrbracket = \llbracket [m] \times [n] \rrbracket = \llbracket [m+n] \rrbracket = \beta_{m+n}.$$

The caret thus *adds* the indices of base multinumbers. In this sense the caret operates on the index structure in the same way that multiplication of polynumber monomials adds exponents — a parallel that connects to the Umbral Calculus of G.C. Rota, where formal operations on polynomial indices encode combinatorial identities. Note also that $\beta_0 = \alpha = [1]$ is the caret identity: $\beta_0 \wedge A = [1] \wedge A = [1 \times a : a \in A] = A$ for any box A .

Example 10.2 Here is a caret of more general multinumbers involving

$B = [0, [3], [2, 4]] = 1 + \alpha_3 + \beta_2\alpha_4$ and $C = \llbracket [1, 1], [2, 4] \rrbracket = \beta_1 + \beta_2\alpha_4$:

$$\begin{aligned} B \wedge C &= [0, [3], [2, 4]] \wedge \llbracket [1, 1], [2, 4] \rrbracket \\ &= [0 \times [1, 1], 0 \times [2, 4], [3] \times [1, 1], [3] \times [2, 4], [2, 4] \times [1, 1], [2, 4] \times [2, 4]] \\ &= [0, 0, [4, 4], [5, 7], [3, 3, 5, 5], [4, 6, 6, 8]] \\ &= 2 + \alpha_4^2 + \alpha_5\alpha_7 + \alpha_3^2\alpha_5^2 + \alpha_4\alpha_6^2\alpha_8. \end{aligned}$$

Example 10.3 Here are some caret powers of

$M = [A, B] : M \wedge M = [A \times A, A \times B, B \times A, B \times B] = [A^2, {}_2(A \times B), B^2]$ while

$M \wedge M \wedge M =$

$[A \times A \times A, A \times A \times B, A \times B \times A, A \times B \times B, B \times A \times A, B \times A \times B, B \times B \times A, B \times B \times B] :$
 $[A^3, {}_3(A^2 \times B), {}_3(A \times B^2), B^3].$

The Fundamental Identity of Arithmetic

11.1 Summation for multinumbers

The summation recipe is also interesting for multinumbers and the caret operation. If q is a multinumber, say $q = [r_0, r_1, \dots, r_l]$ for polynumbers $r_0, r_1, \dots, r_l$ then $\Sigma(q) = r_0 + r_1 + \dots + r_l$ is a polynumber.

Proposition 24 For multinumbers $r, s, \dots, w$ we have the **Caret product identity**:

$$\Sigma(r \wedge s \wedge \dots \wedge w) = \Sigma(r) \times \Sigma(s) \times \dots \times \Sigma(w).$$

Proof. An element of $r \wedge s \wedge \dots \wedge w$ has the form $a \times b \times \dots \times l$ with $a \in r, b \in s, \dots, l \in w$. Summing over all such product tuples

$$\sum_{a \in r, b \in s, \dots, l \in w} a \times b \times \dots \times l = \left(\sum_{a \in r} a \right) \times \left(\sum_{b \in s} b \right) \times \dots \times \left(\sum_{l \in w} l \right) = \Sigma(r) \times \Sigma(s) \times \dots \times \Sigma(w)$$

This multiplicativity of Σ under the caret is the box-arithmetic analogue of the Euler product formula, where multiplicative functions on natural numbers factor over prime contributions.

This plays an interesting role in reformulating certain classical analytic number theory identities into statements in (finite) box arithmetic. In particular we have the following Fundamental Identity of Arithmetic

Theorem 3 (Fundamental Identity of Arithmetic (FIA)) Suppose that n is a natural number and that we have, in order, all of the prime numbers $2, 3, 5, \dots, p$ less than or equal to n . Then

$$\rho(2)_n \wedge \rho(3)_n \wedge \dots \wedge \rho(p)_n =_n \zeta_n.$$

Explicitly this is the identity

$$\left[1, 2, 2^2, \dots, 2^{m(2,n)}\right] \wedge \left[1, 3, 3^2, \dots, 3^{m(3,n)}\right] \wedge \dots \wedge \left[1, p, p^2, \dots, p^{m(p,n)}\right] =_n [1, 2, 3, \dots, n]$$

a caret product over all the prime power *-polynumbers $\rho(q)_n$ where q runs through the primes from 2 to p .

Proof. The elements of the caret product are all possible products $2^{a_1} \times 3^{a_2} \times 5^{a_3} \times \dots \times p^{a_k}$ where $0 \leq a_i \leq m(q_i, n)$ where q_i is the i -th prime in between 2 and p . By the Fundamental Theorem of Arithmetic, every positive integer up to n has a unique factorisation into prime powers; hence the elements of the caret product that do not exceed n are exactly the integers $1, 2, 3, \dots, n$, each appearing exactly once. This is precisely the polynumber $[1, 2, 3, \dots, n]$, establishing the identity up to degree n .

The FIA is the box-arithmetic form of two classical results simultaneously: the Fundamental Theorem of Arithmetic (unique prime factorisation) and the Euler product formula, which expresses the Riemann zeta function as a product over primes: so we could write

$$\sum_{k=1}^n \alpha^k =_n \prod_{p \leq n} \left(1 + \alpha^p + \alpha^{p^2} + \dots + \alpha^{p^{m(p,n)}}\right).$$

Here the box-arithmetic version works entirely over the natural numbers, with no complex analysis, no convergence questions, and no infinite products — the identity is exact and finite at each level n .

Example 11.1 If $n = 12$ then the primes less than or equal to 12 are 2, 3, 5, 7, 11, and the identity is

$$[1, 2, 4, 8] \wedge [1, 3, 9] \wedge [1, 5] \wedge [1, 7] \wedge [1, 11] =_{12} [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12] \text{ or}$$

$$(\alpha + \alpha^2 + \alpha^4 + \alpha^8) \wedge (\alpha + \alpha^3 + \alpha^9) \wedge (\alpha + \alpha^5) \wedge (\alpha + \alpha^7) \wedge (\alpha + \alpha^{11}) =_{12} (\alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 +$$

11.2 Productation and caretation recipes

Just as the summation recipe Σ takes a box M and adds its elements, we now introduce two further recipes following the same pattern: productation Π , which

multiplies the elements of M , and caretation Υ which carets them. These three recipes — Σ, Π, Υ — correspond respectively to the first three levels of the operation hierarchy, each collecting elements under the operation at its level.

Definition The **product recipe** Π takes boxes to boxes, and is defined as follows: if $M = \lfloor A, B, \dots, K \rfloor$ is a non-zero box, then the **production of** M is

$$\Pi[M] \equiv A \times B \times \dots \times K$$

and also we define $\Pi(0) \equiv 0$.

Definition The **caret recipe** Υ takes boxes to boxes, and is defined as follows: if $M = \lfloor A, B, \dots, K \rfloor$ is a non-zero box, then the **caretation of** M is

$$\Upsilon[M] \equiv A \wedge B \wedge \dots \wedge K$$

and also we define $\Upsilon(0) \equiv 0$.

Similarly for the production we have $\Pi[M + N] = \Pi[M] \times \Pi[N]$ and $\Pi[\lambda \times M] = (\Pi(M))^\lambda$.

Example 11.2 If $M = \lfloor A, B \rfloor$ and $\lambda = 3$ then

$$\Pi[3 \times M] = \Pi[\lfloor A, A, A, B, B, B \rfloor] = A^3 \times B^3 = (A \times B)^3 = (\Pi(M))^3.$$

Similarly for the caretation we have $\Upsilon[M + N] = \Upsilon[M] \wedge \Upsilon[N]$ and $\Upsilon[\lambda \times M] = \lambda^{s[M]} \times \Upsilon[M]$.

The operation hierarchy and arithmetic template

The arithmetical operations of Box Arithmetic form a hierarchy, and this hierarchy carries with it a corresponding hierarchy of order relations, divisibility theories, and fundamental theorems. We develop this unified structure now, as it organises much of the material so far in the book, but at the cost of considerable generalization and abstraction.

12.1 The operation hierarchy

We write $\times_n$ for the n -th operation in the hierarchy, so that

$$\times_0 \equiv + \quad \times_1 \equiv \times \quad \times_2 \equiv \wedge \quad \times_3, \times_4, \dots$$

with each operation generated from the previous by the uniform schema: the successor of $\times_n$ is the operation $\times_{n+1}$ defined by collecting all possible $\times_n$ -combinations of elements:

$$A \times_{n+1} B \times_{n+1} \dots \times_{n+1} K \equiv [a \times_n b \times_n \dots \times_n k : a \in A, b \in B, \dots, k \in K].$$

The standard symbols $+$ and $\times$ are retained throughout the book; the subscripted forms $\times_n$ appear when the hierarchical structure is being emphasised.

Every operation $\times_n$ with $n \geq 1$ distributes over $\times_0 = +$:

$$A \times_n (B + C) = (A \times_n B) + (A \times_n C)$$

since collecting all $\times_{n-1}$ -combinations with elements from $B + C$ is the same as collecting separately from B and C . The pair $(\times_0, \times_n)$ therefore forms a commutative semiring at each level: $\times_0$ is addition and $\times_n$ is multiplication.

However $\times_n$ does *not* in general distribute over $\times_m$ for $1 \leq m < n$. For example, caret does not distribute over multiplication for general polynumbers: with $A = [1, 2]$ and $B = C = \alpha$, we have $A \wedge (B \times C) = \alpha^2 + \alpha^4$ while $(A \wedge B) \times (A \wedge C) = \alpha^2 + 2\alpha^3 + \alpha^4$. Partial distributivity holds on monomials α^m , where the identity $m(p + q) = mp + mq$ at the exponent level gives the required identity. The full distributivity story between non-adjacent operations is part of the unexplored algebra of the higher levels.

12.2 The level filtration

The inductive definition of boxes generates a natural hierarchy. A box M is a **level- $(n + 1)$ box** for $n \geq 0$ if every element of M is a level- n box, with the base case that the empty box 0 is the unique level-0 box. Equivalently:

- **Level-0:** the empty box 0 alone.
- **Level-1:** boxes of level-0 elements — the natural numbers Nat
- **Level-2:** boxes of level-1 elements — the polynumbers Poly
- **Level-3:** boxes of level-2 elements — the multinumbers Multi
- **Level- n :** boxes of level- $(n - 1)$ elements.

Since every level- $(n - 1)$ box is automatically level- n , we obtain a genuine filtration:

$$\{0\} = \mathbf{Level-0} < \mathbf{Level-1} < \mathbf{Level-2} < \mathbf{Level-3} < \cdots$$

The **level** $\ell(M)$ of a box M is the smallest n for which M is a level- n box. This hierarchy is *filtered*, not *graded*: each level contains all previous levels. The empty box 0 has level 0; every other natural number has level 1; the polynumber $\alpha = [1]$ has level 2; the base multinumber $\beta_1 = [\alpha]$ has level 3.

Proposition 25 (Identity elements at each level) The identity element for the operation $\times_n$ is a level- n box. It is unique and we denote it e_n . Then

$$e_0 = 0 \quad e_1 = 1 = [0] \quad e_2 = \alpha = [1] \quad e_3 = \beta_1 = [\alpha] \quad \dots$$

with $e_{n+1} = [e_n]$ for each n .

Proof. Each identity is the box containing the previous identity as its sole element: $0 \rightarrow [0] = 1 \rightarrow [1] = \alpha \rightarrow [\alpha] = \beta_1 \rightarrow \cdots$ Each step increases the level by exactly 1. That each e_n is indeed the identity for $\times_n$ follows by checking that $e_n \times_n A = A$ for

any box A , using the fact that e_n has exactly one element e_{n-1} , and $e_{n-1} \times_{n-1} a = a$ for any element $a \in A$ by the inductive hypothesis.

Proposition 26 (Operations preserve level) If M has level at most m and N has level at most n , then $M \times_k N$ has level at most $\max(m, n)$ for any k .

Proof. Elements of $M \times_k N$ have the form $a \times_{k-1} b$ with $a \in M$ and $b \in N$. Since a and b have levels at most $m - 1$ and $n - 1$ respectively, and any operation applied to lower-level objects stays at or below that level, $a \times_{k-1} b$ has level at most $\max(m, n) - 1$. Hence $M \times_k N$ has level at most $\max(m, n)$.

12.3 The characteristic operation theorem

The hierarchy of operations and the level filtration are the same structure viewed from two perspectives.

Theorem 4 (Characteristic operations) For level- n boxes A and B , we have $A \times_k B = A \times_n B$ for all $k \geq n$. The operation $\times_n$ is thus the **characteristic operation** of level n : the first operation to act in a genuinely new way at that level, with all higher operations collapsing to it.

Proof. By induction on n . For $n = 0$: A and B are both the empty box 0 , and $0 \times_k 0 = [a \times_{k-1} b : a \in 0, b \in 0] = 0$ for all k . For the inductive step: let A and B be level- n boxes, so their elements $a \in A$ and $b \in B$ are level- $(n - 1)$ boxes. For $k \geq n$:

$$A \times_k B = [a \times_{k-1} b : a \in A, b \in B].$$

Since $k - 1 \geq n - 1$, the inductive hypothesis gives $a \times_{k-1} b = a \times_{n-1} b$ for all elements a, b . Hence $A \times_k B = [a \times_{n-1} b : a \in A, b \in B] = A \times_n B$.

Corollary 5 Caret on natural numbers equals multiplication: $m \wedge n = m \times n$ for all $m, n \in \mathbb{N}$. All operations $\times_k$ with $k \geq 2$ agree with caret on polynumbers.

Proof. Natural numbers are level-1 boxes, so by the theorem all operations $\times_k$ with $k \geq 1$ agree with $\times_1 = \times$ on natural numbers. Since $m \times n$ gives ordinary multiplication of natural numbers, so does $m \wedge n$. Similarly polynumbers are level-2 boxes, so all $\times_k$ with $k \geq 2$ agree with $\times_2 = \wedge$ on polynumbers.

12.4 The hierarchy of order relations

Each operation $\times_n$ induces a natural partial order on boxes.

Definition For boxes A and B , write $A \leq_n B$ if there exists a box C such that $A \times_n C = B$. The minimum element of $\leq_n$ is the identity e_n .

The three levels already developed in the book give:

- $\leq_0$: multiset inclusion — $A \leq_0 B$ iff $A + C = B$ for some C (Definition ?? of Chapter ??).
- $\leq_1$: multiplicative divisibility — $A \leq_1 B$ iff $A \times C = B$ for some C (Section ?? of Chapter ??).
- $\leq_2$: caret divisibility — $A \leq_2 B$ iff $A \wedge C = B$ for some C .

By the Characteristic Operation Theorem, $\leq_k$ on level- n boxes reduces to $\leq_n$ for all $k \geq n$: the deepest meaningful order on level- n boxes is $\leq_n$.

12.5 The arithmetic template

The divisibility theory — factors, primes, irreducibles, GCD, LCM, and the Fundamental Theorem of Arithmetic — is an abstract theory that applies to any operation satisfying appropriate conditions. We state it once, then apply it at every level.

Arithmetic template Let $\star$ be an operation on boxes with identity e , satisfying commutativity, associativity, **cancellation** ($A \star B = A \star C$ with $A \neq e$ implies $B = C$), and the **identity-product property** ($A \star B = e$ implies $A = B = e$). Then define:

- A is a **$\star$ -factor** of B if $A \leq_\star B$.
- A is **$\star$ -irreducible** if $A \neq e$ and $A = B \star C$ implies $B = e$ or $C = e$.
- A is **$\star$ -prime** if $A \neq e$ and $A \leq_\star B \star C$ implies $A \leq_\star B$ or $A \leq_\star C$.
- $\gcd_\star(A, B)$: the $\leq_\star$ -greatest common lower bound of A and B .
- $\text{lcm}_\star(A, B)$: the $\leq_\star$ -least common upper bound of A and B .
- The **$\star$ -FTA**: every element factors uniquely into $\star$ -irreducibles.

Proposition 27 Each operation $\times_n$ satisfies the conditions of the arithmetic template on level- n boxes, with identity e_n . Hence the full divisibility theory — $\times_n$ -factors, $\times_n$ -primes, $\times_n$ -irreducibles, $\gcd_n$, lcm_n , and the $\times_n$ -FTA — is defined and meaningful at every level n .

Proof. Commutativity and associativity hold at every level by the general properties of the operation hierarchy. Cancellation and the identity-product property follow by induction on n , using Propositions ?? and ?? of Chapter ?? as base cases for level 1.

12.6 The GCD-LCM identity at every level

Theorem 6 (GCD-LCM identity) For any boxes A and B at level n :

$$\gcd_n(A, B) \times_n \text{lcm}_n(A, B) = A \times_n B.$$

Proof. At level 0 this is the inclusion-exclusion identity $(A \cap B) + (A \cup B) = A + B$ of Proposition ?? in Chapter ?. At level 1 it is $\gcd(m, n) \times \text{lcm}(m, n) = m \times n$ for natural numbers, proved in Section ?. At each higher level: $\gcd_n$ takes the $\times_{n-1}$ -minimum and lcm_n takes the $\times_{n-1}$ -maximum of the $\times_n$ -exponents in the level- n factorisation, and the identity $\min(a, b) + \max(a, b) = a + b$ (Proposition ?? of Chapter ??) applied at the exponent level gives the result.

Remark 4 This single identity unifies three classical results. At level 0 it is the inclusion-exclusion principle. At level 1 it is the GCD-LCM product formula. At level 2 and above it is a new family of identities, one at each level of the hierarchy. The Proposition 18 lattice identities — idempotent, commutative, associative, absorption, and distributive laws — hold for $(\gcd_n, \text{lcm}_n)$ at every level n , with $(\gcd_0, \text{lcm}_0) = (\min, \max)$ on natural numbers being the first and simplest instance.

12.7 The arithmetic at each level

We record the three fully developed levels.

Level 0 ($\times_0 = +$, identity 0): The $\times_0$ -irreducibles are the single-element boxes $[A]$ for any box A . The $\times_0$ -FTA says that every box decomposes uniquely as a sum of its elements — this is simply the definition of a box as a multiset. The $\gcd_0$ is intersection $\cap$ and the lcm_0 is union $\cup$, and $(A \cap B) + (A \cup B) = A + B$.

Level 1 ($\times_1 = \times$, identity 1): The $\times_1$ -irreducibles among natural numbers are the classical prime numbers. The $\times_1$ -FTA is the classical Fundamental Theorem of Arithmetic. The $\gcd_1$ and lcm_1 are the classical greatest common divisor and least common multiple, and $\gcd(m, n) \times \text{lcm}(m, n) = m \times n$.

Level 2 ($\times_2 = \wedge$, identity α): For monomials, $\alpha^m \wedge \alpha^n = \alpha^{mn}$, so $\times_2$ -factorisation of α^n corresponds exactly to $\times_1$ -factorisation of the exponent n . The $\times_2$ -irreducibles among monomials are α^p for classical primes p . For general polynumbers the $\times_2$ -irreducibles are those whose element multisets admit no non-trivial multiplicative splitting; binomials $\alpha^m + \alpha^n$ are $\times_2$ -irreducible precisely when $\gcd_1(m, n) = 1$.

12.8 The Fundamental Theorem at every level

Theorem 7 (FTA at every level) The $\times_n$ -FTA holds at every level n : every level- n box factors uniquely into $\times_n$ -irreducibles.

Proof. By induction on n , using the Cashwell-Everett theorem at each step. *Cashwell and Everett (1959)* proved that if R is a UFD then the ring $R[[x_1, x_2, \dots]]$ of formal power series in countably many variables over R is also a UFD. Setting $R_0 = \mathbb{Z}$ (a UFD) and defining inductively

$$R_n \equiv R_{n-1}[[x_P : P \in \mathcal{P}_{n-1}]]$$

where $\mathcal{P}_{n-1}$ is the countable set of level- n irreducibles, each R_n is a UFD by Cashwell-Everett applied to R_{n-1} . The ring R_n is the ring of formal Dirichlet series at level n , and $\times_n$ -factorisation of level- n boxes corresponds to factorisation in R_n . Since R_n is a UFD, factorisation is unique. The restriction from $\mathbb{Z}$ -coefficients to $\mathbb{N}$ -coefficients is valid because products of polynomials with all positive coefficients must have positive coefficient factors — negative coefficients cannot cancel in a product that yields only positive coefficients.

Remark 5 This proof strategy is uniform across all levels: one application of the Cashwell-Everett theorem per level. However the concrete content grows in complexity at each step: identifying the irreducibles at level n requires full understanding of the factorisation theory at level $n - 1$. The complete development of the level- n arithmetic for $n \geq 3$ is a substantial programme, portions of which we leave for subsequent work.

12.9 The two-dimensional arithmetic grid

Classical number theory — divisibility, primes, GCD, LCM, the Fundamental Theorem of Arithmetic — is, from the perspective of Box Arithmetic, the arithmetic of a single

cell: $\times_1$ -irreducibility among level-1 boxes. Box Arithmetic reveals this to be one entry in an infinite two-dimensional grid of arithmetic theories, indexed by pairs (n, k) with $0 \leq k \leq n$: the theory of $\times_k$ -irreducibility among level- n boxes. By the Characteristic Operation Theorem, $\times_k$ on level- n boxes reduces to $\times_n$ for $k > n$, so only the region $k \leq n$ gives genuinely distinct theories.

Level $n \setminus \text{Op } \times_k$	$\times_0$	$\times_1$	$\times_2$	$\times_3$
1 ($\mathbb{N}$)	trivial	classical primes	$= \times_1$	$= \times_1$
2 (polynumbers)	trivial	poly irred. over $\mathbb{N}$	Dirichlet poly irred.	$= \times_2$
3 (multinumbers)	trivial	?	?	level-3 irred.
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Each cell (n, k) below and on the diagonal carries its own notions of divisibility, primality, irreducibility, GCD, LCM, and Fundamental Theorem, all instances of the same abstract arithmetic template, all provable via the FTA lifting theorem. The cells below the diagonal — $\times_k$ -irreducibility at level $n > k$ — are perhaps the most intriguing: they ask about lower-level factorisation of higher-level objects. For instance, the $\times_1$ -irreducibles among polynumbers (cell $(2, 1)$) are the polynomials irreducible over $\mathbb{N}[\alpha]$ under ordinary multiplication — a theory connecting polynomial algebra to classical number theory. Classical number theory, polynomial irreducibility theory, Dirichlet series theory, and many new theories not yet named are all cells in this grid. Exploring even a few of these appears to be a rich mathematical programme.

Theorem 8 (FTA lifting) The $\times_n$ -FTA at level n follows from the $\times_{n-1}$ -FTA at level $n - 1$ via the characteristic operation alignment: $\times_n$ -irreducibles at level n correspond to $\times_{n-1}$ -primes at level $n - 1$ through the index structure of level- n objects.

Proof. At level 2: $\times_2$ -factorisation of monomials α^n corresponds to $\times_1$ -factorisation of the exponent n , so $\times_2$ -irreducible monomials α^p correspond to $\times_1$ -primes p . At each higher level the same correspondence holds: the index structure of level- n objects encodes the level- $(n - 1)$ arithmetic, and factorisation at level n reduces to factorisation at level $n - 1$ through this encoding.

This principle reveals the tower of Fundamental Theorems not as a collection of separate results but as a single theorem at every level, each lifting the previous through the characteristic operation. The entire arithmetic of Box Arithmetic — from the simplest decomposition of a multiset into its elements, through the classical theory of prime numbers, to the caret arithmetic of polynomials and beyond — is one unified abstract structure, repeated and enriched at each level of the hierarchy.

12.10 The collection recipe hierarchy

Just as the operations $\times_n$ form a hierarchy, so do the recipes that collect the elements of a box under each operation. The summation recipe Σ , the productation recipe Π , and the caretation recipe Υ introduced in Chapter ?? are the first three instances of a single unified family.

Definition For each natural number n , the **level- n collection recipe** Π_n takes boxes to boxes and is defined as follows: for a non-zero box $M = [A, B, \dots, K]$, the **level- n collection of** M is

$$\Pi_n [M] \equiv A \times_n B \times_n \dots \times_n K$$

the $\times_n$ -product of all elements of M , and we define $\Pi_n [0] \equiv e_n$, the identity element for $\times_n$.

The existing collection recipes are recovered immediately: $\Pi_0 = \Sigma$ (summation), $\Pi_1 = \Pi$ (productation), and $\Pi_2 = \Upsilon$ (caretation). Note that $\Pi_n [0] = e_n$ since an empty $\times_n$ -product returns the identity, and $\Pi_n [[A]] = A$ since the $\times_n$ -product of a single element is that element.

Example 12.1 We have $\Pi_0 [[3, 3, 4]] = 3 + 3 + 4 = 10$, $\Pi_1 [[3, 3, 4]] = 3 \times 3 \times 4 = 36$, and $\Pi_2 [[\alpha, \alpha^2, \alpha^3]] = \alpha \wedge \alpha^2 \wedge \alpha^3 = \alpha^{1 \times 2 \times 3} = \alpha^6$.

The fundamental identities

Proposition 28 (Additive multiplicativity) For any boxes M and N and any natural number n :

$$\Pi_n [M + N] = \Pi_n [M] \times_n \Pi_n [N].$$

Proof. The box $M + N$ contains all elements of M together with all elements of N . Taking the $\times_n$ -product of all of them gives $\Pi_n [M] \times_n \Pi_n [N]$, since $\times_n$ is commutative and associative.

This single proposition encapsulates three results already in the book: for $n = 0$ it says $\Sigma [M + N] = \Sigma [M] + \Sigma [N]$, for $n = 1$ it says $\Pi [M + N] = \Pi [M] \times \Pi [N]$, and for $n = 2$ it says $\Upsilon [M + N] = \Upsilon [M] \wedge \Upsilon [N]$.

Proposition 29 (Scalar power rule) For any box M , any natural number n , and any natural number λ :

$$\Pi_n [\lambda \times M] = (\Pi_n [M])^{\times_n \lambda}$$

where $A^{\times_n \lambda}$ denotes the λ -fold $\times_n$ -product of A with itself.

Proof. By induction on λ , using additive multiplicativity:

$$\begin{aligned} \Pi_n [\lambda \times M] &= \Pi_n [(\lambda - 1) \times M + M] = \Pi_n [(\lambda - 1) \times M] \times_n \Pi_n [M] = \\ &= (\Pi_n [M])^{\times_n (\lambda - 1)} \times_n \Pi_n [M] = (\Pi_n [M])^{\times_n \lambda}. \end{aligned}$$

Remark 6 The scalar power rule unifies three familiar results. For $n = 0$ it gives $\Sigma [\lambda M] = \lambda \times \Sigma [M]$ (linearity of Σ). For $n = 1$ it gives $\Pi [\lambda M] = (\Pi [M])^\lambda$ (the power rule for production already in Chapter ??). For $n = 2$ it gives $\Upsilon [\lambda M] = (\Upsilon [M])^{\wedge \lambda}$ (the λ -fold caret power of the caretation). In each case the natural number scalar λ becomes a $\times_n$ -exponent.

The product rule at each level

Theorem 9 (Product rule at level n) For any boxes M and N and any natural number n :

$$\Pi_n [M \times_{n+1} N] = (\Pi_n [M])^{\times_n s[N]} \times_n (\Pi_n [N])^{\times_n s[M]}.$$

Proof. Elements of $M \times_{n+1} N$ are of the form $a \times_n b$ with $a \in M$ and $b \in N$. Taking the $\times_n$ -product of all such elements:

$$\begin{aligned} \Pi_n [M \times_{n+1} N] &= \times_n\text{-product of all } a \times_n b \\ &= \times_n\text{-product over } a \text{ of } (\times_n\text{-product over } b \text{ of } a \times_n b) \\ &= \times_n\text{-product over } a \text{ of } a^{\times_n s[N]} \\ &= (\times_n\text{-product over } a \text{ of } a)^{\times_n s[N]} \times_n (\times_n\text{-product over } b \text{ of } b)^{\times_n s[M]} \end{aligned}$$

which gives $(\Pi_n [M])^{\times_n s[N]} \times_n (\Pi_n [N])^{\times_n s[M]}.$

Remark 7 The product rule at level n unifies two results already in the book. For $n = 0$ it gives $\Pi_0 [M \times N] = s[N] \times \Pi_0 [M] + s[M] \times \Pi_0 [N]$, the Leibniz product rule for Σ from Chapter ??. For $n = 1$ it gives $\Pi_1 [M \wedge N] = (\Pi_1 [M])^{s[N]} \times (\Pi_1 [N])^{s[M]}$, a power-weighted product rule for Π under the caret. This level-1 identity does not appear explicitly in the book before now. The pattern continues at every level: the product rule at level n is the Leibniz rule of Box Arithmetic, expressed in the arithmetic of level n .

The cross-level identity

The Caret Product Identity of Chapter ?? —

$\Pi_0[r \wedge s \wedge \cdots \wedge w] = \Pi_0[r] \times \Pi_0[s] \times \cdots \times \Pi_0[w]$ — is a special case of a more general family of identities, one for each pair of levels.

Theorem 10 (Cross-level identity) For any boxes M and N , any natural number n , and any $k \geq n + 2$:

$$\Pi_n[M \times_k N] = \Pi_n[M] \times_{k-1} \Pi_n[N].$$

Proof. Elements of $M \times_k N$ are of the form $a \times_{k-1} b$ with $a \in M$ and $b \in N$. So

$$\Pi_n[M \times_k N] = \times_n\text{-product of all } a \times_{k-1} b.$$

Since $k \geq n + 2$ we have $k - 1 \geq n + 1 > n$, so $\times_{k-1}$ distributes over $\times_n$ (as every operation distributes over all lower operations). Therefore:

$$\times_n\text{-product of all } a \times_{k-1} b = (\times_n\text{-product over } a) \times_{k-1} (\times_n\text{-product over } b) = \Pi_n[M] \times_{k-1} \Pi_n[N].$$

Remark 8 The condition $k \geq n + 2$ is essential. When $k = n + 1$ the identity fails in this form and is replaced by the product rule of the previous subsection. When $k = n$ we would be computing $\Pi_n[M \times_n N]$, a still more complex expression. The cross-level identity thus operates in the region strictly above the diagonal of the two-dimensional arithmetic grid.

Corollary 11 (Iterated cross-level identity) For boxes $M_1, M_2, \dots, M_r$ and any $k \geq n + 2$:

$$\Pi_n[M_1 \times_k M_2 \times_k \cdots \times_k M_r] = \Pi_n[M_1] \times_{k-1} \Pi_n[M_2] \times_{k-1} \cdots \times_{k-1} \Pi_n[M_r].$$

Proof. By induction on r , applying the cross-level identity at each step.

Corollary 12 (Caret Product Identity) The Caret Product Identity of Chapter ?? is the special case $n = 0, k = 2$ of the cross-level identity:

$$\Pi_0[r \wedge s \wedge \cdots \wedge w] = \Pi_0[r] \times \Pi_0[s] \times \cdots \times \Pi_0[w].$$

The full table of identities

The identities above may be arranged by the levels n and k involved. The entry in position (n, k) records the identity for $\Pi_n[M \times_k N]$.

$n \setminus k$	$k = n$	$k = n + 1$	$k = n + 2$	$k \geq n + 3$
any n	complex	Product rule	Cross-level	Cross-level
$n = 0$	Σ of $+$	Leibniz rule	Caret product id.	$\Pi_0[M] \times_{k-1} \Pi_0[N]$
$n = 1$	Π of $\times$	Power product rule	level-1 cross-level	$\Pi_1[M] \times_{k-1} \Pi_1[N]$
$n = 2$	Υ of $\wedge$	level-2 product rule	level-2 cross-level	$\Pi_2[M] \times_{k-1} \Pi_2[N]$

Connection to the derivative and the FIA

For polynumbers, the collection recipe Π_0 connects to the derivative recipe D by the identity $\Pi_0[p] = D[p](1)$, established in Chapter ??: the level-0 collection of a polynumber equals its derivative evaluated at 1. The cross-level identity then gives a further connection: applying Π_0 to a caret product of polynumbers converts it into an ordinary product of their derivatives at 1.

The Fundamental Identity of Arithmetic (Theorem ?? of Chapter ??) is the iterated cross-level identity at $n = 0$, $k = 2$, applied to the prime tower polynumbers:

$$\Pi_0[\rho(2)_n \wedge \rho(3)_n \wedge \cdots \wedge \rho(p)_n] = \Pi_0[\rho(2)_n] \times \Pi_0[\rho(3)_n] \times \cdots \times \Pi_0[\rho(p)_n]$$

where the right hand side is the Euler product expansion, and the left hand side equals $\Pi_0[\zeta_n] = 1 + 2 + 3 + \cdots + n$ by the FIA itself. The cross-level identity is thus the algebraic engine driving the FIA.

Remark 9 The collection recipe hierarchy Π_n and the operation hierarchy $\times_n$ are in a precise sense dual to each other: $\times_n$ builds up boxes from their elements, while Π_n collapses boxes back down to a single value by combining all elements under $\times_n$. The identities in this section record how these two hierarchies interact across levels, and the cross-level identity in particular shows that Π_n converts a $\times_k$ -product of boxes into a $\times_{k-1}$ -product of values, stepping down one level in the operation hierarchy. This downward stepping is the algebraic mechanism underlying both the Euler product formula and the Fundamental Identity of Arithmetic.